UNIVERSITY OF COPENHAGEN
Department of Nordic Studies and Linguistics (NorS)

Master Thesis

Helga Ingrun Maria Thurn

Cross-lingual Representation and Explainability of Named Entities in Multilingual Transformer Models

Supervisor: Ali Basirat

Date: 30/05/2026

Name of university:	University of Copenhagen
Name of faculty:	Faculty of Humanities
Name of department:	Department of Nordic Studies and Linguistics (NorS)
Author:	Helga Ingrun Maria Thurn
Title and subtitle:	Cross-lingual Representation and Explainability of Named Entities in Multilingual Transformer Models
Topic description:	This thesis investigates how multilingual Transformer models represent named entities across languages, focusing on whether entity representations are organised more strongly by semantic type than by language and how this relates to model architecture and attribution behaviour.
Supervisor:	Ali Basirat
Date:	30 May 2026
Number of characters:	135817
Number of standard pages:	56,6

Abstract

Multilingual Transformer models achieve strong performance on cross-lingual named entity recognition (NER), but benchmark scores alone do not explain how named entity representations are internally organised across languages. This thesis investigates whether multilingual NER models form partially shared semantic spaces for named entities or whether their representations remain primarily shaped by language. The study uses English, Spanish, and Dutch CoNLL NER data and compares two multilingual Transformer paradigms: XLM-R as a sequence-tagging model and mT5 as a generative text-to-text model. The analysis focuses on span-level entity representations across languages, entity types, layers, and model architectures. Entity embeddings are extracted from trained models and examined using representation-geometric methods, including k-nearest-neighbour purity, silhouette scores, clustering recovery, centroid distances, and two-dimensional projections. An exploratory attribution analysis based on Integrated Gradients is also included to examine whether predictions rely mainly on entity-local evidence or on surrounding context. The results show that named entity representations are organised more strongly by entity type than by language, especially in XLM-R. XLM-R develops clear semantic clustering in deeper layers, while language-based separation remains comparatively weak but not absent. mT5 shows a similar overall tendency, but with weaker geometric structure. The analyses further indicate that entity categories differ in their degree of cross-lingual organisation, and that residual language-sensitive structure remains visible in local neighbourhood patterns. The exploratory attribution results suggest that predictions are influenced not only by the target entity span but also by surrounding context. Overall, the findings suggest that multilingual NER models develop partially shared semantic spaces for named entities, while these spaces remain shaped by architecture, layer depth, entity type, and residual language-specific variation. The thesis contributes a span-level perspective on multilingual representation geometry and its relation to multilingual NER behaviour.

The project repository is available on GitHub:

https://github.com/mariathurn/multilingual-ner-thesis

Table of contents

ABSTRACT 3

INTRODUCTION 6

THEORETICAL & CONCEPTUAL BACKGROUND 8

Named Entity Recognition 8

Transformer Models 11

Cross-lingual Representation Learning 13

Representation Geometry in NLP 15

Explainability & Attribution 16

Related Work 17

METHODOLOGY 19

Data 19

Preprocessing pipeline 20

Tokenization 20

Subword alignment 20

Span extraction 21

Maximum sequence length 21

Models 22

XLM-R tagging model 22

Generative model 22

Embedding Extraction 24

Geometry Metrics 25

PCA 26

UMAP 26

Silhouette Score 27

KMeans Clustering with NMI and ARI 27

kNN Purity 28

Centroid Distance 28

Summary 29

Explainability and Error Analysis **29**
Attribution Analysis 29
Error Taxonomy 30

Experimental Design **31**

RESULTS **34**

Predictive performance **34**

Clustering by language and entity type **34**

Layer-wise evolution **38**

Entity-type differences in clustering **42**

Model-paradigm differences **44**

Exploratory attribution analysis **45**

DISCUSSION **47**

Summary of Key Findings **47**

Why does semantic alignment increase across layers? **48**

Why is XLM-R more entity-oriented than mT5? **48**

Theoretical Implications **49**

Comparison to Prior Work **52**

Limitations **54**

CONCLUSION **56**

BIBLIOGRAPHY **58**

APPENDIX **66**

Introduction

Multilingual natural language processing has become an increasingly important area of research as language technologies are deployed in settings that extend far beyond monolingual English use (Hedderich et al., 2021; Qin et al., 2025). Many practical applications of language technology, including information extraction, search, question answering, and digital public services, must function across languages in real-world settings (European Commission, 2024; Pai et al., 2024). At the same time, multilingual systems must contend with substantial differences in resource availability, writing systems, and sociolinguistic conditions across languages (Doğruöz & Sitaram, 2022; Hedderich et al., 2021).

In this broader landscape, multilingual models are often expected to support transfer across languages by learning internal representations that capture shared meaning rather than merely language-specific surface form (Hämmerl et al., 2024; Mousi et al., 2024). Yet this expectation raises a fundamental question: to what extent do multilingual models actually construct shared semantic spaces, and where do language-specific structures persist? This question is especially significant for multilingual named entity recognition (NER), where a successful cross-lingual transfer requires models to capture shared entity meaning while at the same time accounting for language-specific differences in how entities are written and used (Shah et al., 2024; Yadav et al., 2024).

NER is a particularly suitable test case for studying multilingual representation learning because it requires models to recognise and classify referential expressions such as persons, locations, organisations, and miscellaneous named entities within running text (J. Li et al., 2022; Sang & De Meulder, 2003). Named entities often have strong semantic regularities that may generalise across languages. A person's name, for example, may denote the same type of entity regardless of whether it appears in English, Dutch, or Spanish. However, named entities are also highly sensitive to language-specific orthography, morphology, tokenisation, and contextual conventions. Organisation names may be translated, partially adapted, or embedded in local naming practices; location references may vary in form across languages; and annotation boundaries may interact with language-dependent syntax (Jain et al., 2019; Xie et al., 2018). NER therefore provides a useful lens through which to study the tension between semantic abstraction and language-specific structure in multilingual Transformer models.

Recent multilingual Transformer models such as XLM-R are commonly understood as learning cross-lingual representations that support transfer between languages, particularly in

settings where supervised data are limited for some languages (Conneau, Khandelwal, et al., 2020; Ruder et al., 2019). The underlying assumption is that hidden representations encode semantic content in a form that can be shared across languages, allowing information learned from one language to benefit another. Although this transfer is often evaluated through downstream task performance, such performance does not fully explain how transfer is internally realised. Two models may obtain similar task scores while relying on different representational strategies, and apparent transfer success may conceal substantial residual language separation in the hidden space. For this reason, analysing internal representation geometry is important. Rather than focusing exclusively on end-task performance, a geometric perspective asks how examples are organised in representation space: whether they cluster by semantic category, by language, by both, or by neither. Such analysis can clarify whether multilingual models develop partially shared semantic representations across languages, or whether they instead retain largely language-specific internal structure despite strong downstream performance.

Prior work has examined multilingual representation geometry and transfer in multilingual NER, but comparatively less attention has been paid to how named entity span representations are organised across languages and entity types (Chang et al., 2022; Mueller et al., 2020). This leaves open important questions: if multilingual models create partially shared spaces for named entities, are those spaces primarily structured by entity type, does this structure emerge across layers, and do different model paradigms show similar or divergent geometric patterns? These questions are relevant for interpreting model-internal organisation and for understanding how multilingual generalisation is reflected in representation space.

This thesis addresses those questions by examining cross-lingual representation geometry and explainability in multilingual NER across English, Dutch, and Spanish. More specifically, it investigates how named entity representations are organised across layers in multilingual Transformer models and to what extent they form shared cross-lingual semantic spaces rather than language-specific clusters. The core problem can be formulated as a tension between two possible organising principles. One possibility is that entity representations cluster primarily by language, suggesting that multilingual transfer remains constrained by language-specific encoding. Another is that they cluster more strongly by semantic type, indicating the emergence of partially language-invariant semantic structure. A third possibility is that both forces coexist, with semantic grouping and language separation interacting to different degrees depending on layer, entity category, and model setup.

To investigate how such organisation differs across modelling conditions, the thesis compares two paradigms that occupy different positions in contemporary NER research: sequence-

tagging models and generative models. Sequence tagging represents the conventional formulation of NER, where a model predicts token-level labels over an input sequence (J. Li et al., 2022). Generative approaches instead reformulate the task as text generation, requiring the model to produce structured outputs or target sequences that encode entity information (Yan et al., 2021). These paradigms differ not only in output format, but potentially also in how they learn and organise internal representations. Comparing them therefore provides a useful opportunity to examine whether differences between the two modelling setups are reflected in cross-lingual entity geometry.

For this comparison, the thesis uses XLM-R as the sequence-tagging model and mT5 as the generative model. XLM-R was selected because it is a widely used multilingual encoder model for cross-lingual sequence labelling, while mT5 was selected because it provides a multilingual encoder–decoder framework suitable for text-to-text NER. This choice allows the thesis to compare two established multilingual Transformer setups that differ in task formulation and architecture while being trained and analysed on the same multilingual NER data.

Theoretical & Conceptual Background

Named Entity Recognition

NER is the task of identifying textual spans that refer to predefined categories of real-world entities, such as persons, organisations, locations, or miscellaneous named expressions (Nadeau & Sekine, 2007; Sang & De Meulder, 2003). In practical terms, an NER system must both detect the boundaries of an entity mention and assign it the correct semantic label. Because named entities play an important role in information extraction, question answering, search, and knowledge base construction, NER has become a standard task in natural language processing and a common benchmark for studying linguistic generalisation across languages (Nadeau & Sekine, 2007).

A central reason why NER is relevant in multilingual research is that it combines semantic regularities with language-specific variation. Entity categories such as person, location, and organisation are conceptually meaningful across languages, yet the way they are written and used in context can differ substantially (Mueller et al., 2020). Because orthography, morphology, word order, and naming conventions vary across linguistic contexts, multilingual NER provides a useful setting for examining whether multilingual models learn representations that reflect shared semantic structure, language-specific form, or a combination of both (Mueller et al., 2020; Pires et al., 2019).

For this reason, NER is particularly suitable for studying cross-lingual representation geometry in multilingual Transformer models.

One of the most established formulations of NER is the sequence tagging paradigm, in which a sentence is treated as a sequence of tokens and the model predicts a label for each token (Sang & De Meulder, 2003). The most common annotation scheme is the BIO tagging scheme, where labels encode both entity type and token position within an entity span (Ramshaw & Marcus, 1995; Sang & De Meulder, 2003). A token marked with B indicates the beginning of an entity span, I indicates that the token continues an entity span of the same type, and O indicates that the token is outside any named entity span (Ramshaw & Marcus, 1995; Sang & De Meulder, 2003). Under this scheme, a phrase such as *New York City* may be represented as B-LOC, I-LOC, I-LOC, where LOC denotes a location entity, while non-entity tokens receive the label O. BIO tagging provides a simple and effective way to convert span detection into token-level classification while preserving information about entity boundaries (Ramshaw & Marcus, 1995).

The sequence tagging paradigm has been dominant in NER for many years because it maps naturally onto supervised token classification (Nadeau & Sekine, 2007; Sang & De Meulder, 2003). Conceptually, the model learns contextual token representations and assigns each token a label based on its surrounding context. The resulting predictions can then be combined into entity spans according to the BIO structure. This paradigm is especially well suited to datasets such as CoNLL-style corpora, where annotations are already formatted as token-level BIO labels (Sang & De Meulder, 2003). In multilingual settings, sequence tagging models are commonly used to investigate cross-lingual transfer in NER, and they are particularly suitable for the present thesis because their token-level predictions make it straightforward to derive span-level entity representations from contextual token embeddings (Mueller et al., 2020; Rahimi et al., 2019).

An alternative formulation is the generative NER paradigm, in which NER is cast not as token-level classification but as a text generation task. Instead of assigning a label to each token, the model is prompted to generate a structured textual output that expresses the entities found in the input. For example, given a sentence, a generative model may produce a sequence specifying the extracted entities and their types in a predefined format. Conceptually, this reframes NER as conditional sequence generation: the model must interpret the sentence, identify relevant spans, and render the result in the target output format (Yan et al., 2021).

This generative view differs from sequence tagging in several important ways. First, it does not require prediction to be tied directly to the input token sequence on a one-label-per-token basis. Second, it relies on the model's ability to represent both the input sentence and the output schema within the same sequence-to-sequence framework. Third, it offers a more flexible formulation that

can be extended to related structured prediction tasks (Yan et al., 2021). At the same time, this flexibility introduces different representational demands: whereas sequence tagging emphasises token-local decisions under contextual encoding, generative NER requires the model to transform sentence understanding into a coherent output sequence. This distinction is particularly relevant in the present thesis, since architectural differences between tagging and generative models may be reflected in the internal organisation of named entity representations.

Although sequence tagging and generative NER differ in formulation, both aim to recover the same underlying information: the boundaries and semantic types of entity mentions in text. Their comparison is therefore theoretically useful. If two paradigms achieve similar task performance while organising internal representations differently, this may reveal important differences in how multilingual models encode entity meaning and language structure. For a thesis concerned with cross-lingual representation geometry, NER is thus not only an applied task but also an analytical lens on multilingual representation learning.

NER systems are typically evaluated using precision, recall, and F1-score, most often at the span-level rather than the token level (Nadeau & Sekine, 2007; Sang & De Meulder, 2003). Span-level evaluation requires that a predicted entity match the gold annotation in both boundary and type to count as correct. Precision measures the proportion of predicted entities that are correct, recall measures the proportion of gold entities that are successfully recovered, and F1 summarises the balance between the two. This evaluation setup is important because token-level accuracy alone can be misleading: a model may label many individual tokens correctly while still failing to recover complete entity spans. For NER, the central unit of interest is therefore the entity mention as a structured span, not the isolated token.

In multilingual NER, span-level F1 is particularly appropriate because it evaluates whether models recover complete entity mentions across languages, rather than only assigning correct labels to isolated tokens. This is important in cross-lingual settings, where transfer performance can vary across languages and entity types (Mueller et al., 2020; Rahimi et al., 2019). A high score suggests that the model is not only identifying entity-related tokens but also assembling them into correct semantic units. However, performance metrics alone do not explain how this is achieved internally. Two models may obtain similar F1 scores while relying on different representational strategies. This motivates the broader perspective of the present thesis: beyond evaluating whether models succeed on multilingual NER, it is necessary to examine how named entity information is internally organised across layers, languages, and modelling paradigms.

Transformer Models

Transformer models provide the architectural basis for most contemporary multilingual language models. Introduced by Vaswani et al. (2017), the Transformer replaces recurrent processing with stacked attention and feed-forward layers, allowing tokens in a sequence to interact more directly and in parallel. In this architecture, each token representation is updated through repeated layers, producing a sequence of contextual hidden states whose content changes as information is integrated across the network. This layered structure is central to later analyses of representation geometry, since the hidden state of a token at a given layer can be interpreted as the model's intermediate representation of that token in context (Vaswani et al., 2017).

A key component of the Transformer is self-attention. In self-attention, each position in a sequence computes a weighted interaction with other positions in the same sequence in order to form a context-sensitive representation. Vaswani et al. (2017) define self-attention as a mechanism that relates different positions of a single sequence to compute a representation of that sequence, implemented through query, key, and value projections combined by scaled dot-product attention. Because each token can attend to potentially all other tokens, self-attention enables the model to capture long-range dependencies without the step-by-step recurrence used in earlier architectures, making the model both more parallelisable and more flexible in how contextual information is combined (Vaswani et al., 2017).

Transformers are built from multiple stacked layers, and the hidden states across layers are not simply repetitions of the same information. Rather, different layers tend to encode different aspects of linguistic structure. Bidirectional Encoder Representations from Transformers (BERT), for example, is explicitly designed to pre-train deep bidirectional representations from unlabelled text by conditioning on both left and right context in all layers (Devlin et al., 2019). Subsequent analyses suggest that these contextual representations are distributed across the network in a structured way, with lower and middle layers often capturing more local or syntactic information and higher layers contributing more strongly to semantic or task-relevant abstractions, even if this progression is not perfectly discrete (Jawahar et al., 2019; Tenney et al., 2019). This is relevant since any claim about cross-lingual clustering or representational geometry depends on which layer is being examined: multilingual structure is not fixed at a single representational level but emerges through the model's hierarchy of hidden states.

Another core property of Transformer language models is their reliance on large-scale pretraining objectives. In BERT, pretraining is based primarily on masked language modelling, where some input tokens are masked and the model learns to predict them from context; the original BERT setup also includes next sentence prediction, an auxiliary objective that trains the model to

decide whether two sentences occur consecutively in the input text (Devlin et al., 2019). More generally, such objectives encourage the model to learn contextual regularities from raw text before being fine-tuned for downstream tasks. As a result, downstream performance does not rely on task-specific supervision alone, but builds on a pre-existing representational system already shaped by broad distributional patterns in language (Devlin et al., 2019). In multilingual models, this pretraining stage is particularly important because it is where cross-lingual regularities may begin to emerge, even before any supervised transfer task is introduced. In multilingual settings, Transformer models are often pretrained on large text collections from many languages. Multilingual BERT (mBERT) showed that cross-lingual transfer can emerge even without explicit alignment supervision, and XLM-R later showed that large-scale multilingual masked language modelling can produce strong cross-lingual transfer across many tasks (Conneau, Khandelwal, et al., 2020; Pires et al., 2019). Together, these findings suggest that joint multilingual pretraining can support cross-lingual generalisation.

This leads to the key conceptual question for multilingual representation analysis: why might multilingual pretraining produce shared semantic spaces? One reason is that a single model with shared parameters processes multiple languages using the same representational machinery. When similar meanings occur in partially comparable contexts across languages, the model may learn representations that abstract away from purely language-specific surface forms. A shared subword vocabulary can further support interaction across languages, especially where scripts, morphemes, or lexical fragments overlap. At the same time, shared vocabulary alone cannot fully explain cross-lingual transfer, since mBERT has been shown to transfer even between languages with different scripts and little or no lexical overlap (Pires et al., 2019). This suggests that shared multilingual representations may arise because the same model parameters are trained across multiple languages, using common pretraining objectives and partially comparable distributional patterns across languages (Conneau, Khandelwal, et al., 2020; Pires et al., 2019).

However, these shared spaces are not perfectly language-neutral. Pires et al. (2019) show that mBERT develops multilingual representations, but also that transfer works better for some language pairs than for others and is influenced by factors such as typological similarity and lexical overlap. XLM-R similarly demonstrates strong multilingual transfer, but not evidence of a completely uniform cross-lingual representation space (Conneau, Khandelwal, et al., 2020). For this reason, multilingual Transformer models are best understood not as producing fully unified semantic spaces, but as learning representational spaces in which shared semantic organisation coexists with residual language-specific structure. This tension is central to the present thesis, which

examines how shared semantic organisation and language-specific structure coexist in multilingual named entity representations.

Cross-lingual Representation Learning

Cross-lingual representation learning concerns the extent to which multilingual models encode semantically similar items from different languages in comparable regions of representational space. In earlier work on multilingual embeddings, this was often framed as an alignment problem: words or sentences with similar meanings across languages should be located closer to one another than semantically unrelated items (Schuster et al., 2019; Upadhyay et al., 2016). More recent work extends this idea to contextual multilingual encoders. In this setting, alignment does not only mean that translated word pairs have similar representations. Instead, it refers more broadly to whether representations with similar meanings are organised similarly across languages and whether this shared structure supports zero-shot transfer (Hämmerl et al., 2024). In this sense, cross-lingual alignment is not an end in itself, but a representational condition that may help explain why models fine-tuned on one language can generalise to another (Hämmerl et al., 2024; Muller et al., 2021).

One factor often associated with cross-lingual alignment is the use of shared subword vocabularies. Multilingual Transformer models such as mBERT and XLM-R use a shared subword vocabulary across many languages, which can create cross-lingual anchor points where languages share scripts, morphemes, or lexical material (Pires et al., 2019). However, vocabulary overlap alone cannot fully explain cross-lingual transfer. Prior work shows that multilingual models can still develop cross-lingual structure under limited lexical overlap, suggesting that shared parameters and multilingual pretraining also play an important role (Conneau, Wu, et al., 2020).

A related question is how implicit alignment via masked language modelling arises at all, given that models such as mBERT and XLM-R are trained without explicit word alignment or parallel supervision. This has been one of the central questions in multilingual natural language processing (NLP). Pires et al. (2019) show that mBERT exhibits substantial zero-shot cross-lingual transfer despite being trained only on monolingual corpora, indicating that multilingual pretraining induces more than simple vocabulary memorisation. Conneau, Wu, et al. (2020) argue that multilingual masked language modelling can lead to emergent cross-lingual structure because a single model with shared parameters must process many languages and therefore discovers symmetries that can be reused across them. Chai et al. (2022) further investigate this phenomenon and argue that cross-lingual ability emerges from structural commonalities between languages, especially compositional regularities, rather than from superficial overlap alone. Together, these

findings support the view that masked language modelling can produce partially shared semantic spaces implicitly, even without explicit cross-lingual supervision.

At the same time, multilingual representations are not fully language-neutral. A recurring theme in the literature is the tension between language-agnostic and language-specific information. Libovický et al. (2020) directly investigate the language neutrality of pretrained multilingual representations and argue that multilingual contextual embeddings contain both semantically shared and language-conditioned components. Müller et al. (2021) similarly analyse mBERT and propose that its success in zero-shot transfer can be understood as the interaction of a multilingual encoder with a task-specific language-agnostic predictor. Other studies likewise report evidence that multilingual models contain identifiable language-specific and language-general components rather than a perfectly unified interlingual space (Hämmerl et al., 2024; Tanti et al., 2021). This means that multilingual representations are better understood as partially aligned: enough shared structure exists to support transfer, but residual language-specific organisation remains.

This distinction is important for the present thesis. If multilingual named entity representations were fully language-agnostic, one would expect clustering primarily by semantic type regardless of language. If, however, multilingual models preserve substantial language-specific structure, then entity representations may cluster simultaneously by semantic category and by language. Cross-lingual representation learning is therefore not only a question of whether transfer works, but also of how shared and language-specific structure coexist in the geometry of the model's hidden states. This provides the conceptual basis for analysing how shared and language-specific structure coexist in multilingual entity representations.

Recent work has also argued against interpreting multilingual processing as either fully language-neutral or as straightforward internal translation into a single dominant language. Basirat (2025) proposes that multilingual models can be understood in terms of coordinate, compound, and sub-coordinate learning, and finds that coordinate learning is the dominant pattern, with only limited evidence for sub-coordinate learning. Shani and Basirat (2025) likewise report only weak evidence for language dominance and argue that multilingual processing is better explained by a combination of shared and language-specific representational spaces. Together, these studies support the view that multilingual representations are neither fully unified nor reducible to dominant-language mediation but instead reflect a graded balance between cross-lingual sharing and residual language-specific structure.

Representation Geometry in NLP

The notion of representation geometry in NLP refers to the structure of the spaces in which model representations are distributed. In Transformer models, each token or sequence can be represented as a high-dimensional vector, often called an embedding or hidden state. An embedding space is therefore the geometric space formed by these vectors, where distances, directions, and neighbourhood relations can be interpreted as reflecting aspects of linguistic similarity or difference. Analysing representation geometry therefore makes it possible to ask whether semantic, syntactic, or task-relevant distinctions are reflected in the spatial organisation of model representations (Devlin et al., 2019; Mikolov et al., 2013; Rogers et al., 2020).

Within such spaces, clustering refers to the tendency of representations with shared properties to group together. If representations of items belonging to the same linguistic or semantic class occupy neighbouring regions, this suggests that the distinction is reflected in the organisation of the representation space. In the present thesis, clustering is used to assess whether semantic category or language identity provides a stronger organising principle in the entity embedding space. Geometric clustering does not by itself explain why a model succeeds on a task, but it offers a useful descriptive view of how information is distributed internally (Hewitt & Manning, 2019; Jawahar et al., 2019; Rogers et al., 2020).

Because embedding spaces are high-dimensional, they are often inspected through projections into two dimensions. In this thesis, two projection methods are used: Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP). PCA is a linear method that represents variation through combinations of the original embedding dimensions, making it useful for inspecting broad global structure in a relatively transparent way. UMAP, by contrast, is a nonlinear method that aims to preserve local neighbourhood structure, which can make local grouping patterns more visible. However, both types of projection reduce complex high-dimensional spaces to a simplified visual form. For this reason, the resulting visualisations are treated as heuristic aids to interpretation rather than exact maps of the original representation space (McInnes et al., 2020).

The reason that such geometric analysis is meaningful is that hidden states can encode structured information. In Transformer models, hidden states are not arbitrary intermediate values, but contextual representations shaped by pretraining objectives and repeated interactions across layers (Devlin et al., 2019). A growing body of work shows that these representations contain recoverable information about linguistic phenomena such as syntax, semantics, and coreference (Jawahar et al., 2019; Tenney et al., 2019). At the same time, the distribution of this information is not uniform, but varies across layers and tasks (Jawahar et al., 2019; Rogers et al., 2020). This

means that the geometry of hidden states can be analysed as evidence about what kinds of structure the model has learned to encode internally (Rogers et al., 2020).

Other studies examine the geometry of hidden states more directly by analysing similarity structure, neighbourhood patterns, clustering behaviour, and changes across layers. Jawahar et al. (2019) show that different layers of BERT capture different levels of linguistic structure. This connects to broader work on the internal analysis of pretrained language models, which examines what these models encode and how such information is organised in their representations, rather than focusing only on downstream performance (Rogers et al., 2020). More recent work on sentence embeddings and fine-tuning likewise highlights that geometric properties of representation spaces are closely tied to whether semantic information is readily accessible and how that accessibility changes under task adaptation (B. Li et al., 2020; Zhou & Srikumar, 2022).

For the purposes of this thesis, representation geometry provides the conceptual bridge between multilingual model behaviour and internal organisation. If multilingual Transformers learn shared semantic abstractions, then this should be reflected not only in cross-lingual task transfer but also in the geometry of their hidden states. Conversely, if language-specific structure remains strong, this may appear as language-based separation or incomplete overlap in embedding space. Analysing clustering, neighbourhood relations, and layer-wise changes is therefore a way of investigating how multilingual named entity information is encoded internally, beyond task performance alone. In this sense, representation geometry is not merely a visual supplement to performance analysis, but a central method for understanding the balance between semantic organisation and language-specific structure in multilingual NLP.

Explainability & Attribution

To complement the representation-geometric analysis, this thesis includes a limited explainability component that examines how token-level attribution is distributed across different regions of the input. Integrated Gradients is applied directly to the XLM-R sequence-tagging model and, more cautiously, in a heuristic source-side setup for the mT5 generative model. The analysis compares attribution assigned to the target token, other tokens within the same entity span, and the left and right context. Rather than treating explainability as a standalone objective, the attribution analysis is used as an exploratory behavioural complement to the representation-geometric results.

In NER, attribution analysis is useful because entity predictions may depend on different kinds of evidence. A model may rely mainly on the entity token itself, on other tokens within the same entity span, or on contextual cues surrounding the mention. This distinction is relevant for multilingual NER because surface forms, entity boundaries, and contextual patterns may vary

across languages. Attribution methods therefore provide a way to examine whether model decisions are associated primarily with entity-local evidence or with broader sentence context.

Integrated Gradients is one such attribution method. It estimates feature importance by accumulating gradients along a path from a baseline input to the actual input (Sundararajan et al., 2017). In this thesis, attribution is treated as a complementary perspective on model behaviour rather than as a complete explanation of the underlying representation geometry.

Related Work

Research relevant to the present thesis lies at the intersection of multilingual named entity recognition, cross-lingual representation learning, and the analysis of internal structure in Transformer models. Prior work has shown that multilingual pretrained models can support effective transfer across languages, including for NER, but it has also shown that such transfer is uneven and shaped by both shared and language-specific factors. At the same time, studies of model internals suggest that multilingual representations are neither fully language-neutral nor fully language-bound. The present thesis builds on these strands of work by focusing specifically on the geometry of named entity representations across languages and by comparing sequence-tagging and generative modelling paradigms.

A first relevant line of research concerns multilingual and cross-lingual NER. Shared-task work such as CoNLL-2002 and CoNLL-2003 established language-independent NER as a benchmark problem and provided standard datasets and evaluation conventions that remain influential in later work (Sang, 2002; Sang & De Meulder, 2003). More recent studies have examined cross-lingual transfer for NER using multilingual pretrained encoders. Rahimi et al. (2019), for example, investigate massively multilingual transfer for NER and show that transfer quality differs substantially across source-target configurations. Mueller et al. (2020) further analyse polyglot NER and argue that multilingual transfer depends not only on additional multilingual supervision, but also on how parameters are shared and adapted during fine-tuning. Related work on low-resource cross-lingual NER likewise shows that multilingual encoders can transfer useful entity-related knowledge across languages, while still being sensitive to lexical, typological, and resource differences (Jain et al., 2019; Xie et al., 2018). Taken together, this literature establishes multilingual NER as both a practically important and theoretically informative setting for studying cross-lingual generalisation.

A second line of research focuses more broadly on cross-lingual representation learning in multilingual language models. mBERT and XLM-R have been central to this literature because they demonstrate substantial zero-shot cross-lingual transfer despite being trained primarily with

multilingual masked language modelling rather than explicit alignment objectives (Conneau, Khandelwal, et al., 2020; Pires et al., 2019). Subsequent work has sought to explain why such transfer emerges. Conneau and Wu (2020) argue that cross-lingual structure can arise from shared model parameters even without shared vocabulary, while Chai et al. (2022) suggest that structural commonalities between languages are especially important for the emergence of cross-lingual ability. Other work has analysed multilingual representations more directly in terms of alignment and language neutrality. Libovický et al. (2020) show that multilingual contextual representations are only moderately language-neutral by default, while Muller et al. (2021) argue that mBERT can be understood as combining a multilingual encoder with a task-specific language-agnostic predictor. More recent surveys similarly describe cross-lingual alignment as a matter of degree rather than a binary property and emphasise the trade-off between shared and language-specific information in multilingual models (Hämmerl et al., 2024). This literature is important for the present thesis because it motivates the question of whether named entity representations form genuinely shared multilingual regions or remain partly separated by language.

A third relevant body of work examines the internal organisation of Transformer representations, including layer-wise structure, probing, and geometric analysis. Studies of BERT have shown that hidden states encode recoverable linguistic information and that this information is distributed non-uniformly across layers (Jawahar et al., 2019; Tenney et al., 2019). Broader reviews of representation analysis in pretrained language models likewise emphasise that internal states should not be treated as opaque by-products, but as meaningful representational spaces whose structure can be investigated directly (Rogers et al., 2020). In the multilingual setting, this perspective has been extended to analyses of geometric organisation across languages. Chang et al. (2022) show that multilingual model representations contain both language-sensitive and language-neutral axes, while Shah et al. (2024) examine correlations between geometric properties and cross-lingual transfer performance. Mousi et al. (2024) further investigate alignment in shared cross-lingual spaces, again suggesting that multilingual representations combine shared structure with persistent language-conditioned variation. These studies are especially relevant because they support the idea that representation geometry can reveal aspects of multilingual organisation that are not visible from evaluation scores alone.

A fourth line of work concerns alternative modelling paradigms for NER, particularly the contrast between sequence tagging and generative formulations. Sequence tagging remains the dominant conventional paradigm in NER and is closely tied to BIO-style labelling and token-level contextual encoding (Ramshaw & Marcus, 1995; Tjong Kim Sang & De Meulder, 2003). However, recent work has increasingly reformulated information extraction tasks in generative terms. Raffel

et al. (2020) provide the broader text-to-text framework that motivates treating NLP tasks as conditional generation, while Xue et al. (2021) extend this multilingual generative paradigm through mT5. For NER specifically, Yan et al. (2021) propose a unified generative framework that casts entity recognition as sequence generation rather than token classification. This line of research is particularly relevant to the present thesis because it introduces a comparison not only of model performance but also of representational assumptions: sequence-tagging models and generative models may solve similar tasks while organising internal information in different ways.

Although these strands of prior work provide important foundations, they have rarely been combined in a single study. Existing multilingual NER research has focused mainly on transfer performance, resource settings, and architectural effectiveness. Work on multilingual representation learning has examined alignment, neutrality, and transfer, but usually at a broader level rather than with a specific focus on named entity categories. Research on representation geometry has shown that multilingual hidden states contain interpretable structure, yet relatively little work has examined how named entity representations cluster across languages and layers in relation to semantic type. Likewise, generative NER research has established the feasibility of text-to-text formulations but has only rarely been connected to questions of cross-lingual internal geometry. The present thesis contributes to this literature by combining multilingual NER with representation-geometric analysis and by comparing how sequence-tagging and generative models structure entity representations.

Methodology

Data

This thesis uses multilingual named entity recognition datasets from the CoNLL shared tasks. English is taken from CoNLL-2003, while Spanish and Dutch are taken from CoNLL-2002. All three datasets were accessed through Hugging Face dataset repositories and converted into a shared preprocessing format. Although CoNLL-2003 also includes German, it was not included because it was not openly available within the Hugging Face-based workflow used for this project. The CoNLL-2003 documentation notes that, for copyright reasons, only the annotations are made available, and that reconstructing the German dataset requires access to the ECI Multilingual Text Corpus. The final language set was therefore restricted to English, Spanish, and Dutch.

All datasets use the same BIO tagging scheme with four entity types: PER, ORG, LOC, and MISC. Under this scheme, each token is assigned a label indicating whether it marks the beginning

of an entity span (B), continues an entity span (I), or lies outside any entity span (O). This produces a unified label space across English, Spanish, and Dutch, which supports consistent model training, evaluation, and representation analysis.

The data are used with their standard train/dev/test splits. No resplitting was performed. Keeping the original partitions ensures comparability with prior work and prevents variation introduced by custom sampling or partition design. The development split is used for validation during model training and model selection, while the test split is reserved for final evaluation and downstream embedding analysis.

Preprocessing pipeline

A shared preprocessing pipeline was applied across all datasets in order to ensure comparability between languages and models. The pipeline converts token-level BIO annotations into model-ready inputs while preserving the information required for both supervised training and later span-based analysis.

First, the original CoNLL token sequences and BIO labels were read sentence by sentence. Sentences were preserved as atomic units throughout preprocessing. Each sentence consists of a token list and a corresponding BIO label sequence of equal length. Label names were normalised into a shared label inventory, and consistent label2id / id2label mappings were used across datasets.

Tokenization

Input sentences were tokenized using the tokenizer associated with the respective pretrained model. For sequence-tagging models, tokenization was applied at the subword level while retaining the original word-level annotation structure. Because Transformer tokenizers may split a single word into multiple subword units, the preprocessing pipeline explicitly tracked the mapping between original tokens and resulting subwords.

Tokenization was performed with truncation enabled and fixed maximum sequence lengths. For the XLM-R tagging setup, input sentences were truncated to a maximum length of 128 tokens. For the mT5 generative setup, the maximum input length was also 128 tokens, while the maximum target length was 64 tokens. The same length settings were used consistently within each experimental setup to ensure that training, evaluation, and embedding extraction operated under comparable input constraints.

Subword alignment

Since BIO labels are defined at the original token level, they must be aligned to subword-tokenized input before training sequence-tagging models. This alignment was performed by mapping each

subword token back to its originating word index. For each original word, the label was assigned only to the first subword, while subsequent subwords were masked from loss computation. This prevents a single gold token label from being counted multiple times when a word is split into several subword pieces.

Special tokens introduced by the tokenizer, such as sentence boundary or padding tokens, were also excluded from supervision. As a result, loss was computed only for positions corresponding to the first subword of each original token. This is a common setup for Transformer-based token classification and preserves compatibility between BIO annotations and subword-based encoders.

Span extraction

In addition to token-level labels used for supervised training, entity spans were extracted from BIO sequences for evaluation and representation analysis. Span extraction converts token-level BIO annotations into structured entity mentions defined by start position, end position, and entity type. A span begins at a B-X label and continues through subsequent I-X labels of the same type until the sequence ends or a different label is encountered.

This span representation is used because the core unit of NER evaluation is the complete entity mention rather than the individual token. It also provides the basis for later analyses of named entity embeddings, where representations are grouped and compared at the entity level rather than only at the token level.

Maximum sequence length

A fixed maximum sequence length was applied during preprocessing to ensure compatibility with Transformer models and stable batch processing. For the XLM-R tagging setup, input sentences were truncated to a maximum length of 128 tokens. For the mT5 generative setup, the maximum input length was also 128 tokens, while the maximum target length was 64 tokens. Sequences exceeding these limits were truncated during encoding.

These limits were sufficient for most sentences in the CoNLL datasets, which are generally short relative to the maximum input length supported by the underlying Transformer models. Applying fixed length settings also ensured consistency between training, evaluation, and embedding extraction.

Overall, the preprocessing pipeline was designed to preserve the original annotation structure while producing model-compatible inputs for multilingual sequence tagging, text-to-text generation, and subsequent representation analysis. The result is a unified multilingual dataset setup

with consistent labels, standard splits, aligned subword supervision, and span-based entity representations.

Models

This study compares two multilingual Transformer-based models for named entity recognition: XLM-R as a sequence-tagging model and mT5 as a generative text-to-text model. The comparison is designed to examine whether two established NER paradigms differ in the way they organise multilingual entity representations. XLM-R provides an encoder-based token-classification setup, while mT5 provides an encoder–decoder setup in which NER is formulated as conditional text generation. Both models were trained on the same multilingual dataset consisting of English, Spanish, and Dutch instances. Model fine-tuning and evaluation were conducted in a GPU-enabled Google Colab environment using the Hugging Face Transformers framework.

XLM-R tagging model

The tagging model is based on `xlm-roberta-base` with a token-classification head. The encoder provides contextualised hidden states for the input sequence, and the classification head maps each supervised token position to one label in the BIO tag inventory. Fine-tuning updated both the pretrained encoder parameters and the task-specific classification head.

In this setup, gold BIO labels were aligned to the first subword of each word, while remaining subword positions were excluded from the loss. The model was trained using the training split only. The development split was used for validation during training and model selection, and final performance was reported on the held-out test split. Evaluation was conducted at the entity-span level using precision, recall, and F1, both overall and separately for English, Spanish, and Dutch.

The main fine-tuning configuration was fixed across experiments: the model used the BIO label scheme with the entity types PER, ORG, LOC, and MISC, a maximum sequence length of 128, a batch size of 8, a learning rate of 2e-5, three training epochs, and the AdamW optimizer.

Generative model

The generative model is based on `google/mt5-small`, a multilingual encoder–decoder Transformer. In contrast to the tagging setup, it formulates named entity recognition as a text-to-text generation task rather than as token-level classification. The model receives an input sentence as a prompt and generates a structured output sequence containing the detected entity spans and their entity types. Fine-tuning updated both the pretrained encoder and decoder parameters.

As in the tagging setup, the model was trained on the combined multilingual dataset consisting of English, Spanish, and Dutch instances. Evaluation was performed on the standard development and test splits, with language-specific scores computed separately for English, Spanish, and Dutch in addition to the overall score. Since the model produces generated text rather than token-level labels, the generated outputs had to be parsed back into entity predictions before scoring.

Fine-tuning followed a supervised sequence-to-sequence setup. Each input sentence was converted into an instruction-style prompt. In the default setup, the prompt took the form `extract named entities: {sentence}`. An optional variant including the language name was also implemented, using prompts of the form `extract named entities from this {lang} sentence: {sentence}`.

The gold BIO labels were converted into a linearised entity-level target representation. First, BIO tags were converted into entity spans paired with entity types. Each entity was then rendered as its surface form followed by its type in angle brackets, using the format `entity span <TYPE>`. Multiple entities were separated by the delimiter `|`. For example, a sentence containing the entities *Barack Obama* (PER) and *Paris* (LOC) would be converted to the target sequence `Barack Obama <PER> | Paris <LOC>`. If a sentence contained no entities, the target string was `none`. The model was then optimised to generate the correct target sequence for each input sentence.

Generated strings were parsed by splitting the output on `|` and matching each segment against the expected pattern `entity span <TYPE>`, where `TYPE` had to be one of PER, ORG, LOC, or MISC. Segments that did not match this format were treated as invalid and excluded from the final prediction set. If the model generated `none`, an empty string, or no valid entity segment, the prediction was treated as containing no entities.

The resulting predicted entity–type pairs were compared with the gold entity–type pairs at the entity level. Precision, recall, and F1 were computed from exact matches between predicted and gold pairs. In addition, exact-match accuracy was computed over the full generated target string. Duplicate predicted entities were collapsed before scoring, since evaluation was based on set comparison of predicted and gold entity–type pairs.

The main fine-tuning configuration was fixed across experiments: the model used a maximum input length of 128, a maximum target length of 64, a generation maximum length of 64, a training batch size of 8, an evaluation batch size of 1, a learning rate of 5e-5, three training epochs, and the AdamW optimizer.

Embedding Extraction

To analyse the internal representation geometry of multilingual named entities, contextual embeddings were extracted from the fine-tuned models after training. This step was designed not to evaluate predictive performance directly, but to create entity-level representations for the later geometric analyses. These embeddings provide the basis for comparing neighbourhood structure, cluster separation, and group-level distances across languages, entity types, layers, and model setups.

Because named entities often consist of multiple tokens, the analysis was conducted at the span-level rather than the individual token level. For each annotated entity mention, the hidden states corresponding to all tokens in the entity span were collected from a selected model layer and combined into a single vector using mean pooling. This produced one embedding per entity mention. Mean pooling was chosen because it provides a simple and stable summary of the full span while preserving information from all tokens in the mention. It is also less sensitive than single-token selection to variation in tokenisation length or internal span structure. Since the object of interest in this thesis is the entity mention as a semantic unit, span-level pooling is more appropriate than analysing individual token representations in isolation.

A second design decision concerns layer selection. Transformer models produce contextual representations at every layer, and these layers are known to differ in the type of information they encode. Earlier layers often reflect more local, lexical, or form-based information, while later layers are typically more task-adapted and more strongly shaped by the supervision used during fine-tuning. For this reason, the analysis begins with the last hidden layer. The final layer is the most directly relevant starting point because it is the representation space closest to the task output and therefore the one most likely to reflect the structure that the fine-tuned model uses when performing named entity recognition. If the model has learned language-independent abstractions useful for NER, these should be especially visible in the final task-adapted representations. Later analyses can then compare this with earlier layers to investigate how cross-lingual alignment develops throughout the network.

To compare entity embeddings quantitatively, similarity was measured using cosine distance. Cosine distance is appropriate in this setting because the main interest lies in the relative orientation of vectors rather than their absolute magnitude. In high-dimensional contextual embedding spaces, vector norms may vary for reasons that are not central to semantic comparison, whereas cosine-based measures capture whether two representations point in similar directions. This makes cosine distance a standard and suitable choice for analysing clustering and neighbourhood structure in contextual representation spaces.

Two complementary families of geometric measures were used: k-nearest-neighbour (kNN) analysis and silhouette-based clustering metrics. kNN analysis provides a local view of the embedding space. For each entity representation, its nearest neighbours can be inspected to determine whether local neighbourhoods are dominated by the same entity type, the same language, or both. This is especially useful for testing whether semantic similarity overrides language-specific structure at a local scale. For example, if a PER entity in Spanish is surrounded mainly by PER entities from multiple languages, this suggests cross-lingual semantic alignment. In contrast, if its neighbours are mainly Spanish entities regardless of type, this would indicate stronger language clustering.

Silhouette metrics provide a more global summary of cluster structure. Whereas kNN focuses on local neighbourhood purity, silhouette scores quantify how well each embedding fits within a group compared with embeddings in other groups. This makes them useful for comparing the relative strength of clustering by entity type and language across the full dataset. In this thesis, silhouette analysis is used to assess whether the overall geometry is better explained by semantic category or language identity. Together, kNN and silhouette measures provide complementary evidence: kNN captures fine-grained local structure, while silhouette scores capture broader global separation.

In combination, these design choices support a representation analysis that is both linguistically meaningful and geometrically interpretable. Span-level pooling ensures that the unit of analysis is the full entity mention; last-layer extraction provides an initial view of the most task-adapted representations; cosine distance offers an appropriate similarity measure for contextual embeddings; and the use of both kNN and silhouette metrics enables comparison of local and global structure in the multilingual entity space.

Geometry Metrics

To characterize the structure of the extracted entity embedding space, several complementary geometric metrics were used. These metrics were chosen to capture different aspects of representation geometry, including low-dimensional structure, cluster separability, local neighbourhood composition, and the relative positions of language and entity-type groups. Some metrics are primarily descriptive and exploratory, while others provide quantitative evidence about the degree to which embeddings are organised by semantic category or by language.

PCA

PCA is a linear dimensionality-reduction method that projects high-dimensional embeddings into a lower-dimensional space while preserving as much variance as possible (Jolliffe & Cadima, 2016). Given a set of centred embedding vectors $x_1, \ldots, x_n \in \mathbb{R}^{\mathbb{d}}$, PCA finds orthogonal directions $u_1, u_2, \ldots$ such that the first principal component u_1 captures the largest possible variance in the data, the second captures the largest remaining variance subject to orthogonality, and so on. Formally, the first principal component is the unit vector u_1 that maximizes

$$\mathrm{Var}(Xu_1)$$

subject to $|u_1| = 1$, where X is the matrix of centered embeddings. The projection of an embedding x into a k-dimensional PCA space is then given by its coordinates along the first k principal directions.

In this thesis, PCA is used primarily as a linear projection tool for visualization and exploratory inspection. It provides a simple view of global variance structure and can reveal whether major axes of variation are associated with entity type, language, or other factors. Because PCA is linear, it is especially useful as a baseline projection against which nonlinear methods can be compared.

UMAP

UMAP is a nonlinear dimensionality-reduction method designed to preserve local neighbourhood structure when projecting data into a low-dimensional space (McInnes et al., 2018). Unlike PCA, which captures global variance through linear projections, UMAP aims to maintain the relational structure of nearby points in the original high-dimensional embedding space.

Conceptually, UMAP first constructs a graph in which each point is connected to its nearest neighbours in the original space. It then learns a low-dimensional layout that preserves this fuzzy neighbourhood graph as closely as possible. The result is a projection in which points that are close in the original space tend to remain close in the reduced space, even when the original structure is nonlinear.

In this thesis, UMAP is used as a tool for visualising nonlinear local structure. This is useful because multilingual contextual embeddings may exhibit manifold structure that is not well captured by linear projections. UMAP can therefore provide a useful qualitative view of local grouping patterns that may not be visible in linear PCA projections. Its role here is exploratory rather than inferential: it helps interpret local geometric tendencies that are later quantified using explicit metrics.

Silhouette Score

The silhouette score is a clustering-quality measure that quantifies how well each point fits within its assigned group relative to other groups (Rousseeuw, 1987). For a point i, let:

- $a(i)$ be the average distance from i to all other points in the same cluster,
- $b(i)$ be the smallest average distance from i to points in any other cluster.

The silhouette value for point i is then

$$s(i) = \frac{b(i) - a(i)}{\max\big(a(i), b(i)\big)}$$

where the value i lies in the range $[-1, 1]$. A value close to 1 indicates that the point is much closer to its own cluster than to other clusters, a value near 0 indicates overlap between clusters, and a negative value suggests that the point may be closer to another cluster than to its assigned one.

In this thesis, silhouette scores are used to evaluate clustering with respect to entity type and language. This makes it possible to compare how clearly the embedding space separates semantic categories and language identities. Silhouette analysis therefore provides a global measure of cluster separation across the full dataset.

KMeans Clustering with NMI and ARI

KMeans is an unsupervised clustering algorithm that partitions a set of embeddings into K clusters by minimizing within-cluster variance. Given cluster centroids $\mu_1, \ldots, \mu_K$, KMeans seeks an assignment of each point x_i to a cluster C_k that minimizes

$$\sum_{k=1}^{K} \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

where the resulting clusters can then be compared with known labels such as entity type or language. Two external clustering evaluation metrics are used for this comparison:
Normalized Mutual Information (NMI) measures the amount of shared information between the cluster assignments and the gold labels. Let U denote the clustering assignments and V the gold labels. Then

$$\mathrm{NMI}(U, V) = \frac{2I(U; V)}{H(U) + H(V)}$$

where $I(U; V)$ is mutual information and H denotes entropy. NMI ranges from 0 to 1, where higher values indicate greater agreement.

Adjusted Rand Index (ARI) measures pairwise agreement between the clustering and the gold labels while correcting for chance (Hubert & Arabie, 1985). It compares whether pairs of points are placed in the same or different clusters in both partitions. ARI ranges from -1 to 1,

though in practice values usually lie between 0 and 1, with higher values indicating stronger correspondence.

In this thesis, KMeans combined with NMI and ARI is used to test whether the unsupervised cluster structure of the embedding space aligns more closely with entity type labels or with language labels. This helps determine whether the unsupervised cluster structure corresponds more closely to semantic or language-specific categories.

kNN Purity

The kNN purity measure captures the local consistency of the embedding space by examining the labels of each point's nearest neighbours. For each embedding x_i, its k nearest neighbours are identified according to a chosen distance measure, here cosine distance. Purity is then computed as the proportion of those neighbours that share a given property with the query point, such as the same entity type or the same language.

For example, if $N_k(i)$ is the set of k nearest neighbours of point i, and y_i is its label, then type-based kNN purity can be defined as

$$\text{purity}_\text{k}(i) = \frac{1}{k} \sum_{j \in N_k(i)} 1\left[y_j = y_i\right]$$

where $1\left[y_j = y_i\right]$ equals 1 if the condition in brackets is true and 0 otherwise. Averaging this value across all points yields an overall purity score.

In this thesis, kNN purity is used to analyse local neighbourhood structure. Two versions are especially relevant: purity with respect to entity type and purity with respect to language. High entity-type purity indicates that local neighbourhoods are semantically coherent, while high language purity indicates stronger language-specific grouping. Because kNN purity focuses on local geometry, it complements the more global perspective provided by silhouette scores and cluster-based metrics.

Centroid Distance

Centroid distance measures the distance between the mean vectors of predefined groups in the embedding space. For a group G containing embeddings $x_1, \ldots, x_m$, the centroid is defined as

$$c_G = \frac{1}{m} \sum_{i=1}^{m} x_i$$

where the distance between two groups G and H is then the distance between their centroids, for example:

$$d(c_G, c_H)$$

In this thesis, centroid distances are used to compare the relative positions of groups defined by entity type, language, or their combinations. For example, one can compare the centroid of Spanish PER entities with the centroid of Dutch PER entities or compare the centroid of PER entities with the centroid of ORG entities. Small cross-lingual centroid distances within the same entity type may indicate stronger alignment across languages, whereas large distances suggest residual language separation. Similarly, large distances between entity-type centroids indicate stronger semantic separation.

Centroid distance therefore provides a simple and interpretable summary of group-level geometry. Unlike kNN purity, which captures local neighbourhood composition, centroid analysis focuses on the relative positions of entire groups.

Summary

Taken together, these metrics provide a multi-level description of multilingual entity representation geometry. PCA and UMAP are used for visualisation, with PCA capturing linear global variance structure and UMAP highlighting nonlinear local structure. Silhouette scores and KMeans-based NMI/ARI quantify cluster separation and label alignment. kNN purity measures local neighbourhood consistency, and centroid distance captures group-level separation. Their combined use allows the analysis to compare local, global, and group-level structure in the entity embedding space.

Explainability and Error Analysis

Attribution Analysis

To complement the geometric analysis, this thesis includes a limited attribution-based explainability component. The purpose of this component is not to provide a full explainability benchmark, but to examine which parts of the input contribute most strongly to model predictions and whether these attribution patterns are broadly consistent with the entity-level structure observed in the representation analysis. Integrated Gradients was selected because it provides a principled gradient-based method for estimating how strongly different parts of the input contribute to a model prediction (Sundararajan et al., 2017).

For the XLM-R tagging model, attribution can be computed directly for token-level label predictions. This makes the resulting attribution signal relatively straightforward to interpret, since the explained decision is tied to a specific input token and its predicted entity label. In the implemented analysis, attribution scores were inspected qualitatively for selected correct predictions and misclassifications. In addition, attribution values were aggregated across predefined

token regions, namely the target token, other tokens within the same entity span, and the left and right context. This made it possible to assess whether attribution was concentrated on the entity token itself, distributed across the full entity span, or assigned more strongly to surrounding contextual cues.

For the mT5 generative model, attribution is less direct because predictions are produced on the decoder side rather than as token-aligned source labels. To obtain an interpretable approximation of source-side attribution in this setting, Integrated Gradients was applied to decoder-side predictions, and the resulting source-side scores were aggregated over coarse input regions. This general strategy is consistent with prior work applying gradient-based attribution analysis to encoder-decoder and sequence-to-sequence models (He et al., 2019). In the final workflow, the mT5 analysis used a simple heuristic decomposition based on an anchor token and its surrounding context rather than a stricter span-matching procedure, since the latter proved too unstable for consistent analysis across examples. The mT5 attribution results are therefore treated more cautiously than the tagging results and are used primarily as a descriptive comparison rather than as a fully parallel attribution analysis.

Overall, the attribution analysis serves an interpretive role. It provides a behavioural perspective on the earlier geometry results by showing which parts of the input the models rely on when producing entity predictions. More specifically, it is used to assess whether attribution is concentrated on linguistically relevant tokens and nearby contextual cues, or whether attribution is diffuse and potentially inconsistent with the predicted entity decision. The explainability component is therefore treated as exploratory and secondary to the core representation-geometric analysis rather than as a standalone evaluation objective.

Error Taxonomy

In addition to attribution analysis, this thesis defines an error taxonomy as a conceptual framework for interpreting model behaviour beyond aggregate performance scores. Fine-grained error categories can be useful for understanding the remaining weaknesses of NER systems, since overall evaluation metrics may conceal qualitatively different failure modes (Rueda et al., 2024). In the final thesis, however, the taxonomy is used primarily as an interpretive framework rather than as a standalone quantitative analysis.

The taxonomy distinguishes between several common types of named entity recognition errors. Boundary errors refer to cases in which the model identifies an entity of the correct general kind but predicts an incorrect span, for example by omitting part of a multi-token entity or including surrounding material that should not belong to the entity mention. Label confusions refer to cases in

which the span is detected but assigned the wrong entity type, such as predicting ORG instead of LOC. Missed entities are cases in which a gold entity is not predicted at all. Spurious entities are the reverse case, in which the model predicts an entity where no gold entity exists. These categories provide a useful way of distinguishing between different failure modes rather than collapsing all errors into a single undifferentiated class.

Experimental Design

The experimental design of this thesis is organised around sub-research questions (SRQs). Each SRQ is linked to specific analyses and evaluation measures, ensuring that the experimental work directly addresses the theoretical questions introduced earlier. This structure is intended to ensure methodological coherence: rather than presenting a loose collection of experiments, each analysis is tied to a specific question about multilingual representation geometry and cross-lingual named entity recognition.

SRQ1. *Do entity representations cluster more strongly by semantic type than by language?*

The first sub-question concerns the primary organising principle of the multilingual embedding space. In this experiment, each annotated entity mention is represented by a span-level embedding extracted from the fine-tuned model. The hidden states belonging to the tokens in the entity span are mean-pooled into a single vector, and the resulting entity representations are compared using cosine distance. To address SRQ1, the same embeddings are evaluated under two alternative grouping labels: semantic type and language. This comparison is based on kNN purity, silhouette scores, and KMeans cluster recovery evaluated with NMI and ARI against both entity-type and language labels. Centroid-distance comparisons provide an additional group-level measure of whether entity-type centroids are more separated than language centroids. Together, these metrics capture local neighbourhood structure, global cluster separation, and unsupervised cluster-label correspondence.

To assess the robustness of the main type-over-language comparison, bootstrap resampling is applied to the final-layer kNN and silhouette differences. The empirical sampling distribution is defined over the set of extracted entity-level representations for each model. In other words, each bootstrap sample is drawn with replacement from the observed pool of entity embeddings in the test set, where each item corresponds to one annotated entity mention together with its language and entity-type label. For each bootstrap sample, the type-based and language-based clustering metrics are recomputed, and the difference between the two values is recorded. This procedure is repeated 1,000 times per model, producing an empirical distribution of type-over-language differences. The

resulting 95% confidence intervals and bootstrap probability $p_{\text{boot}}(\Delta \leq 0)$ are used to evaluate whether the observed type-over-language differences are stable under resampling. Together, these analyses operationalise SRQ1 as a comparison between semantic clustering and language-specific clustering, while also testing whether the observed differences are robust across sampled entity representations.

SRQ2. *How does cross-lingual alignment develop across model layers?*

The second sub-question focuses on how representation structure develops across model layers in the two analysed model setups. Rather than focusing the analysis on the final layer alone, embeddings are extracted from multiple layers of both the XLM-R tagging model and the mT5 generative model and compared systematically. This layer-wise comparison makes it possible to examine whether earlier layers show more language-conditioned structure and whether later layers become more strongly organised by entity type after task fine-tuning. Geometric measures, including kNN purity, silhouette scores, and centroid-based comparisons, are applied across layers in both models to trace how multilingual alignment changes with representational depth. In this way, SRQ2 is addressed as a layer-wise analysis of cross-lingual entity geometry within each model setup.

SRQ3. *To what extent does cross-lingual alignment vary across named entity types?*

The third sub-question investigates whether cross-lingual representation geometry differs across the four named entity categories: PER, ORG, LOC, and MISC. This is examined by analysing language-conditioned structure separately within each entity type. For each category, same-language kNN purity is used to measure whether local neighbourhoods remain concentrated by language, while language silhouette scores measure whether entities of the same type form globally separated language groups. Centroid-based comparisons are used as an additional group-level measure of distances between language-specific subgroups within the same entity type. Together, these analyses make it possible to assess whether some entity types show stronger cross-lingual overlap than others, or whether residual language-specific structure persists more strongly for particular categories.

SRQ4. *How do the XLM-R tagging model and the mT5 generative model differ in their cross-lingual entity representation geometry?*

The fourth sub-question introduces the comparative dimension of the thesis by examining how representation geometry differs between the two implemented model setups. The XLM-R tagging model and the mT5 generative model are trained on the same multilingual dataset and compared under broadly parallel conditions. Embedding extraction and geometric analyses are then

applied to both models, allowing comparison of entity-type clustering, language-conditioned neighbourhood structure, layer-wise development, and group-level separation. This makes it possible to assess not only whether the two setups differ, but also how they differ in the strength and form of cross-lingual semantic organisation.

SRQ5. *What attribution patterns accompany the observed representation-geometric differences?*

The fifth sub-question serves as an exploratory behavioural extension of the core representation-geometric analysis. It examines whether the two model setups show similar or different attribution patterns alongside their geometric differences, without treating attribution as evidence of a causal relationship between representation structure and prediction behaviour.

SRQ5 is addressed using Integrated Gradients. For the XLM-R tagging model, Integrated Gradients is applied to selected examples and aggregated attribution summaries to examine which input regions contribute most strongly to token-level predictions. For the mT5 generative model, the same method is used in a simpler heuristic setup that approximates source-side attribution from decoder-side predictions. The XLM-R analysis is therefore treated as the more direct and methodologically stronger component, while the mT5 analysis is used mainly as a descriptive comparison. The analysis compares how attribution mass is distributed across predefined input regions: the target token, other tokens in the same entity span, left context, and right context. In this way, SRQ5 provides an attribution-based reflection on the main representation findings rather than a separate behavioural evaluation.

Table 1 summarises the analytical design of the thesis by mapping each sub-research question to the corresponding empirical analysis. The first four sub-questions form the core representation-geometric analysis, while SRQ5 is included as an exploratory attribution-based extension.

Table 1. Overview of the experimental design and its relation to the sub-research questions

SRQ	Analytical approach
SRQ1	Type-vs-language comparison using kNN purity, silhouette scores, KMeans cluster recovery with NMI/ARI, and centroid distances
SRQ2	Layer-wise comparison of kNN purity, silhouette scores, and centroid distances in XLM-R and mT5
SRQ3	Per-entity-type analysis of language-conditioned kNN purity, language silhouette, and within-type centroid distances
SRQ4	Comparative analysis of cross-lingual entity geometry in the XLM-R tagging and mT5 generative setups
SRQ5	Exploratory Integrated Gradients analysis of attribution patterns across predefined input regions

Note. The table summarises how each sub-research question is addressed in the empirical analysis. SRQ1–SRQ4 form the core representation-geometric analysis, while SRQ5 is treated as an exploratory behavioural extension based on Integrated Gradients attribution. kNN = k-nearest-neighbour; NMI = Normalised Mutual Information; ARI = Adjusted Rand Index.

Results

This chapter presents the empirical findings of the study. It first reports the overall predictive performance of the two models, then examines clustering by language and entity type, layer-wise changes in representation geometry, entity-type-specific patterns, differences between the XLM-R tagging and mT5 generative setups, and finally the exploratory attribution analysis.

Predictive performance

Before turning to the representation-geometric analyses, this section reports the predictive performance of the two models on the multilingual NER task. This provides a task-level baseline for interpreting later differences in internal representation structure. Table 2 summarises the overall test-set performance of the XLM-R tagging model and the mT5 generative model.

Table 2. Overall test-set predictive performance of XLM-R and mT5.

Model	Precision	Recall	F1	Exact match
XLM-R tagging	0.900	0.911	0.905	—
mT5 generative	0.813	0.817	0.815	0.806

Note. Precision, recall, and F1 are reported at the entity level on the multilingual test set. Exact match is reported only for the mT5 generative model, where the model produces a complete target string and the metric measures whether this generated string exactly matches the gold target string. It is not reported for XLM-R because the tagging model predicts token-level BIO labels rather than a complete output string.

Table 2 shows that the XLM-R tagging model achieves higher overall test-set performance than the mT5 generative model. XLM-R obtains higher precision, recall, and F1, with an overall F1 score of 0.905 compared with 0.815 for mT5. In summary, the XLM-R tagging model shows stronger overall predictive performance than the mT5 generative model on the multilingual test set, providing an initial task-level difference between the two model setups.

Clustering by language and entity type

To address SRQ1, final-layer entity embeddings were evaluated with respect to both language-based and entity-type-based clustering. The comparison uses kNN purity and silhouette scores to assess whether the embedding space is organised more strongly by semantic entity category or language identity.

Table 3. Final-layer clustering by language and entity type with bootstrap confidence intervals.

Model	Same-language kNN	Same-type kNN	Δ kNN type-language	95% CI	Language silhouette	Entity-type silhouette	Δ silhouette type-language	95% CI
XLM-R tagging	0.798	0.972	0.155	[0.149, 0.160]	-0.008	0.703	0.711	[0.704, 0.719]
mT5 generative	0.760	0.829	0.061	[0.055, 0.068]	0.017	0.067	0.050	[0.048, 0.052]

Note. Same-language and same-type kNN measure the proportion of nearest neighbours sharing the same language or entity type. Δ values are computed as entity-type clustering minus language clustering. Positive Δ values therefore indicate stronger organisation by entity type than by language. Confidence intervals are based on 1,000 bootstrap samples. For all reported Δ comparisons, the bootstrap probability of Δ ≤ 0 was 0.000.

Table 3 summarises the final-layer clustering metrics and the corresponding bootstrap confidence intervals for the type-over-language differences. In XLM-R, same-type kNN (0.972) is higher than same-language kNN (0.798), with a mean difference of 0.155 and a 95% bootstrap confidence interval of [0.149, 0.160]. The difference is larger for the silhouette scores: entity-type silhouette (0.703) exceeds language silhouette (-0.008), producing a difference of 0.711 with a 95% confidence interval of [0.704, 0.719]. In mT5, the same direction is observed, but with smaller differences. Same-type kNN (0.829) exceeds same-language kNN (0.760), with a difference of 0.061 and a 95% confidence interval of [0.055, 0.068]. Entity-type silhouette (0.067) also exceeds language silhouette (0.017), with a difference of 0.050 and a 95% confidence interval of [0.048, 0.052].

Thus, in both models, the final-layer metrics are higher for entity type than for language, with larger differences in XLM-R than in mT5. A k-sensitivity check with k = 5, 10, 20, and 50 confirmed the same direction of results across all tested neighbourhood sizes. The full results are reported in Appendix Table 9. For XLM-R, the type-over-language difference ranged from 0.129 to 0.293, while for mT5 it ranged from 0.057 to 0.090. Thus, same-type kNN remained higher than same-language kNN in both models, indicating that the stronger entity-type neighbourhood structure was not specific to the main setting of k = 10.

Table 4. Final-layer cluster-recovery and centroid-distance metrics for XLM-R and mT5.

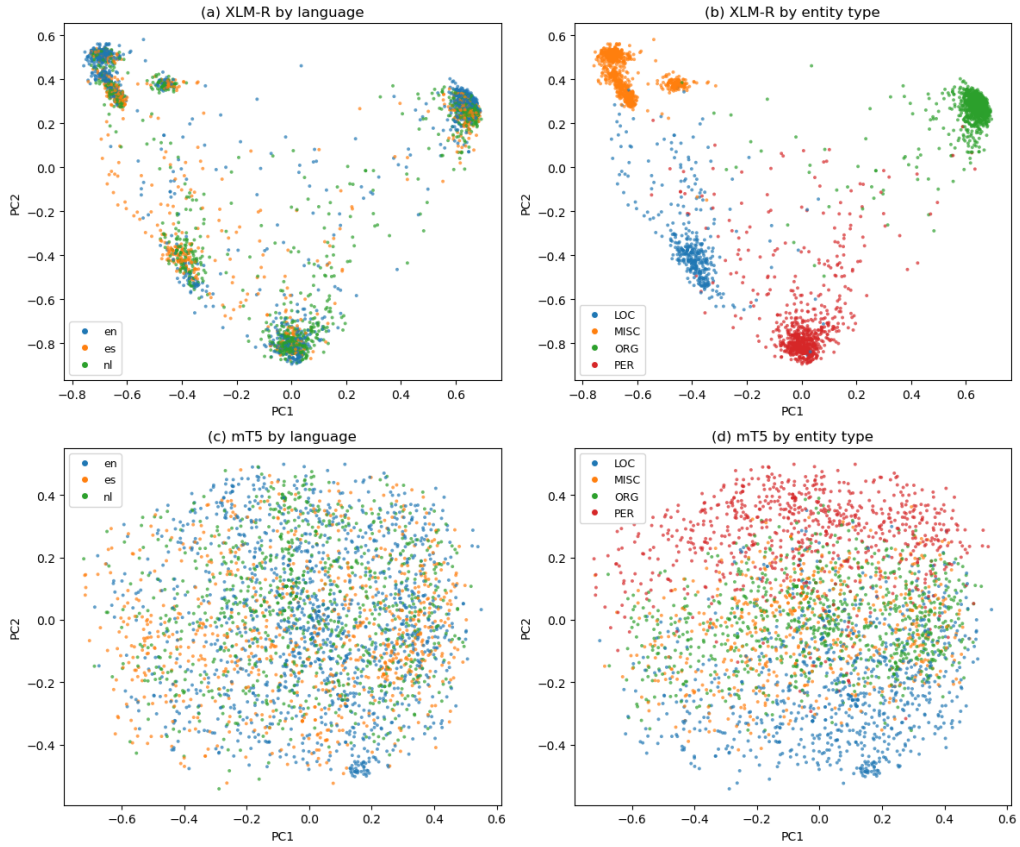
Model	NMI language	NMI entity type	ARI language	ARI entity type	Mean centroid distance language	Mean centroid distance entity type
XLM-R tagging	0.019	0.875	0.023	0.920	0.115	0.997
mT5 generative	0.013	0.108	0.007	0.079	0.077	0.214

Note. NMI = Normalised Mutual Information; ARI = Adjusted Rand Index. Both metrics measure how closely unsupervised KMeans clusters correspond to a reference grouping, here either language or entity type. Values closer to 1 indicate stronger cluster recovery, while values close to 0 indicate little correspondence. Mean centroid distance is computed between group centroids; larger values indicate greater separation between languages or entity types in the representation space.

Table 4 reports the final-layer cluster-recovery and centroid-distance metrics for both models. In XLM-R, KMeans clusters show much stronger agreement with entity type than with language, with entity-type NMI at 0.875 and entity-type ARI at 0.920, compared with language NMI at 0.019 and language ARI at 0.023. This indicates that the unsupervised cluster structure of the final XLM-R representation space closely corresponds to the gold entity categories, but only minimally to language membership. The centroid-distance results support the same pattern: entity-type centroids are much more widely separated from one another than language centroids, with a mean distance of 0.997 compared with 0.115. Together, these results suggest that XLM-R forms a strongly type-oriented representation space in which language differences are present but not the dominant global organising principle.

In mT5, the corresponding entity-type values are lower, with NMI at 0.108 and ARI at 0.079, but they remain higher than the language-based values of 0.013 and 0.007. The mean centroid distance is likewise larger for entity type (0.214) than for language (0.077), although the difference is smaller than in XLM-R. This suggests that mT5 also shows some entity-type organisation, but that this structure is considerably weaker and less recoverable by global clustering. In contrast to XLM-R, the final-layer mT5 representation space appears more diffuse, with less clearly separated entity-type regions. Overall, Table 4 reinforces the conclusion from the kNN and silhouette results: both models organise entity representations more by semantic type than by language, but this tendency is much stronger and more geometrically explicit in the XLM-R tagging model.

Figure 1. Final-layer PCA comparison of XLM-R and mT5 representations coloured by language and entity type.



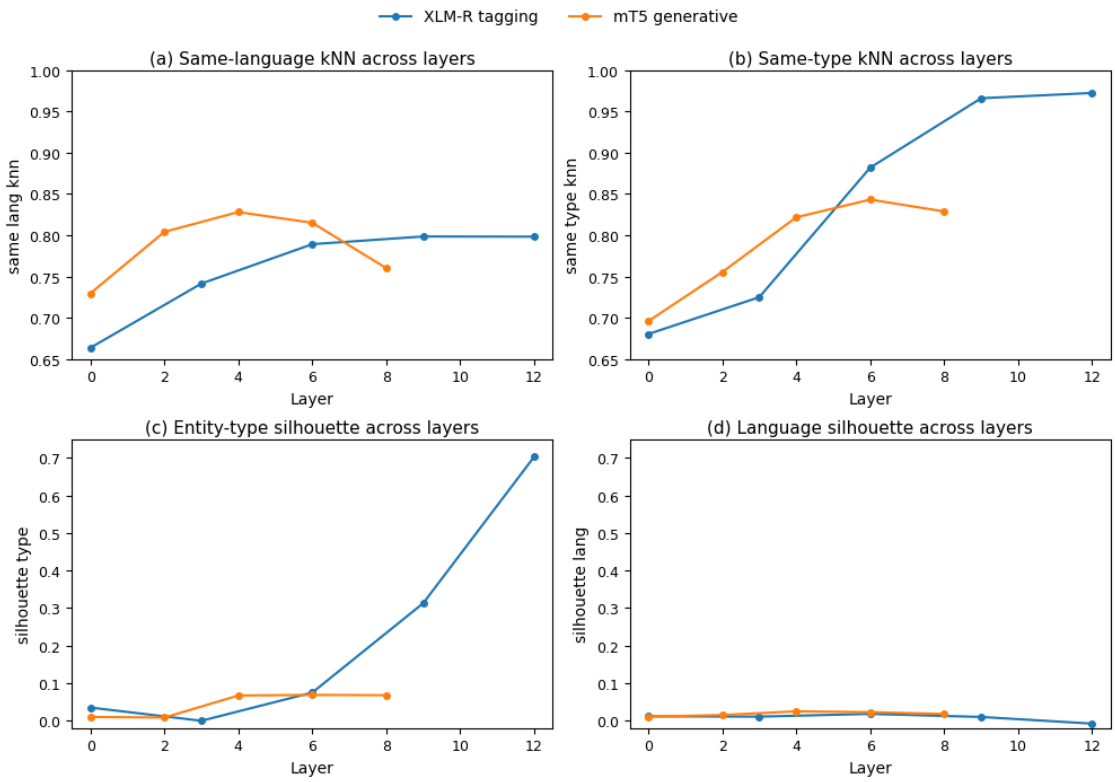
Note. Panel (a) shows the final analysed XLM-R layer coloured by language (English, Spanish, Dutch), and panel (b) shows the same layer coloured by entity type (LOC, MISC, ORG, PER). Panel (c) shows the final analysed mT5 layer coloured by language, and panel (d) shows the same layer coloured by entity type. Entity embeddings were L2-normalised before PCA projection to reduce the influence of differences in vector magnitude between models. PCA was fitted separately for each panel; therefore, the axes should be interpreted as panel-specific visualisation axes rather than directly comparable dimensions.

Figure 1 provides a qualitative comparison of the final analysed representations in the two models. The PCA projections are based on L2-normalised entity representations and therefore emphasise directional structure rather than raw embedding magnitude. In XLM-R, the language-coloured projection shows substantial overlap between languages, whereas the entity-type-coloured projection shows much clearer separation between LOC, MISC, ORG, and PER. In mT5, the language-coloured projection also shows strong overlap, while the entity-type-coloured projection shows a more visible degree of structure. However, the separation remains less sharp and less cleanly partitioned than in XLM-R. Overall, the final-layer results show stronger organisation by entity type than by language in both models. This pattern is visible across neighbourhood-based metrics, silhouette scores, cluster-recovery measures, centroid distances, and the qualitative projections, although the effect is substantially clearer in XLM-R than in mT5.

Layer-wise evolution

This section examines how cross-lingual representation geometry changes across model layers. To address SRQ2, it reports layer-wise trends in neighbourhood purity and silhouette scores for XLM-R and mT5, followed by qualitative PCA and UMAP projections that illustrate the development of language-related and entity-type-related structure across layers.

Figure 2. Layer-wise comparison of cross-lingual representation geometry in XLM-R and mT5.



Note. (a) Proportion of nearest neighbours sharing the same language. (b) Proportion of nearest neighbours sharing the same entity type. (c) Silhouette score with respect to entity type. (d) Silhouette score with respect to language.

Figure 2 shows different layer-wise trends for XLM-R and mT5. In XLM-R, same-type kNN increases strongly across layers, and entity-type silhouette is much higher in the later layers than in the earlier ones. Same-language kNN also increases, while language silhouette remains low across all layers and is slightly negative in the final layer. This suggests that deeper XLM-R layers increasingly organise entity representations according to semantic entity type, while language membership affects local neighbourhoods but does not form globally separated language clusters. In mT5, same-language kNN rises from layer 0 to layer 4 and then decreases, while same-type kNN rises from layer 0 to layer 6 and remains lower than XLM-R in the later layers. Both silhouette measures remain low in mT5 across all analysed layers. This indicates that mT5 develops some local type- and language-related structure, but this structure does not translate into strong global

cluster separation. Overall, the layer-wise results suggest a clearer semantic consolidation across layers in XLM-R than in mT5.

Table 5. Layer-wise cross-lingual representation geometry metrics for XLM-R and mT5.

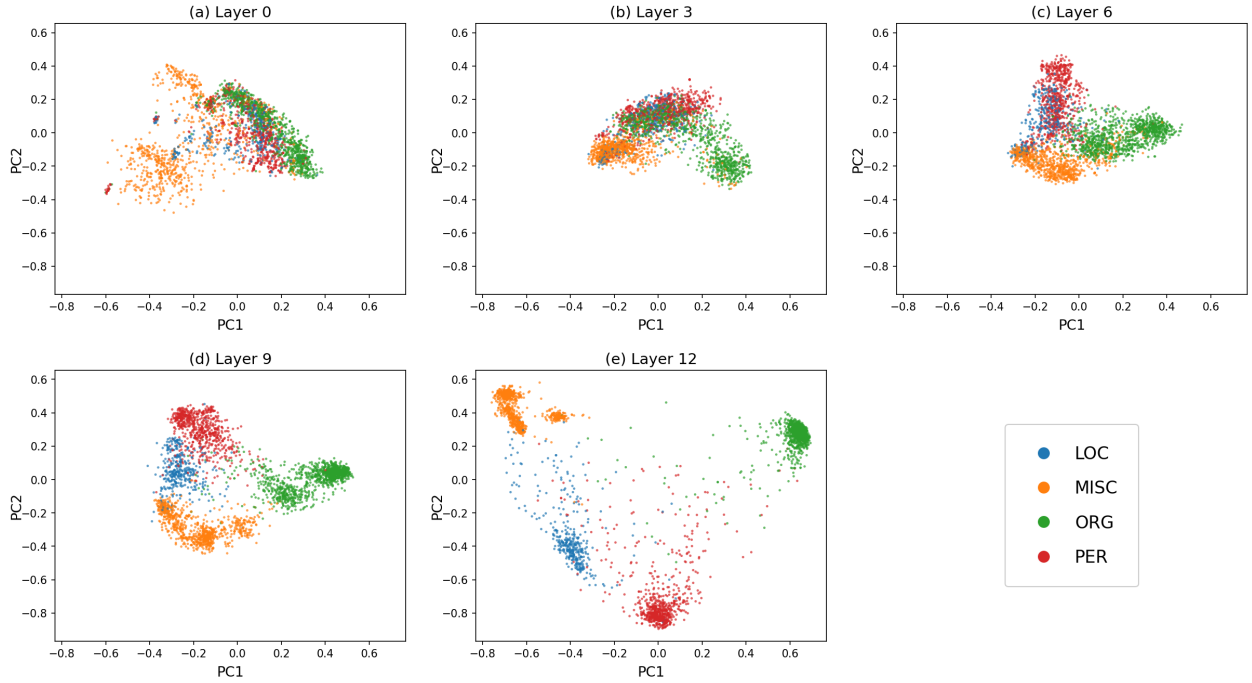
Model	Layer	Same-language kNN	Same-type kNN	Language silhouette	Entity-type silhouette
XLM-R tagging	0	0.664	0.680	0.011	0.035
	3	0.741	0.725	0.010	-0.001
	6	0.789	0.882	0.018	0.074
	9	0.799	0.966	0.009	0.313
	12	0.798	0.972	-0.008	0.703
mT5 generative	0	0.730	0.696	0.010	0.009
	2	0.804	0.755	0.015	0.008
	4	0.828	0.822	0.024	0.067
	6	0.815	0.843	0.022	0.068
	8	0.760	0.829	0.017	0.067

Note. Higher kNN values indicate stronger local neighbourhood concentration, while higher silhouette values indicate clearer separation between groups in the overall representation space.

Table 5 gives the exact values underlying the trends in Figure 2. In XLM-R, same-type kNN increases from 0.680 at layer 0 to 0.972 at layer 12, and entity-type silhouette increases from 0.035 to 0.703. This indicates that entity-type structure becomes progressively stronger across the model, both in local neighbourhoods and in the broader separation of entity-type groups. Same-language kNN also increases, from 0.664 to 0.798, while language silhouette stays close to zero and reaches -0.008 in the final layer. This suggests that language continues to shape local neighbourhood structure but does not produce clearly separated language clusters in the overall representation space.

In mT5, same-language kNN rises from 0.730 at layer 0 to 0.828 at layer 4 and then decreases to 0.760 at layer 8. Same-type kNN increases from 0.696 at layer 0 to 0.843 at layer 6 and is 0.829 at layer 8. Both silhouette measures remain low across all analysed mT5 layers, with language silhouette ranging from 0.010 to 0.024 and entity-type silhouette from 0.008 to 0.068. This suggests that mT5 develops some local type- and language-related structure, but this structure remains weak at the level of global group separation. Overall, the layer-wise metrics indicate a clearer consolidation of entity-type organisation in deeper XLM-R layers than in the analysed mT5 layers.

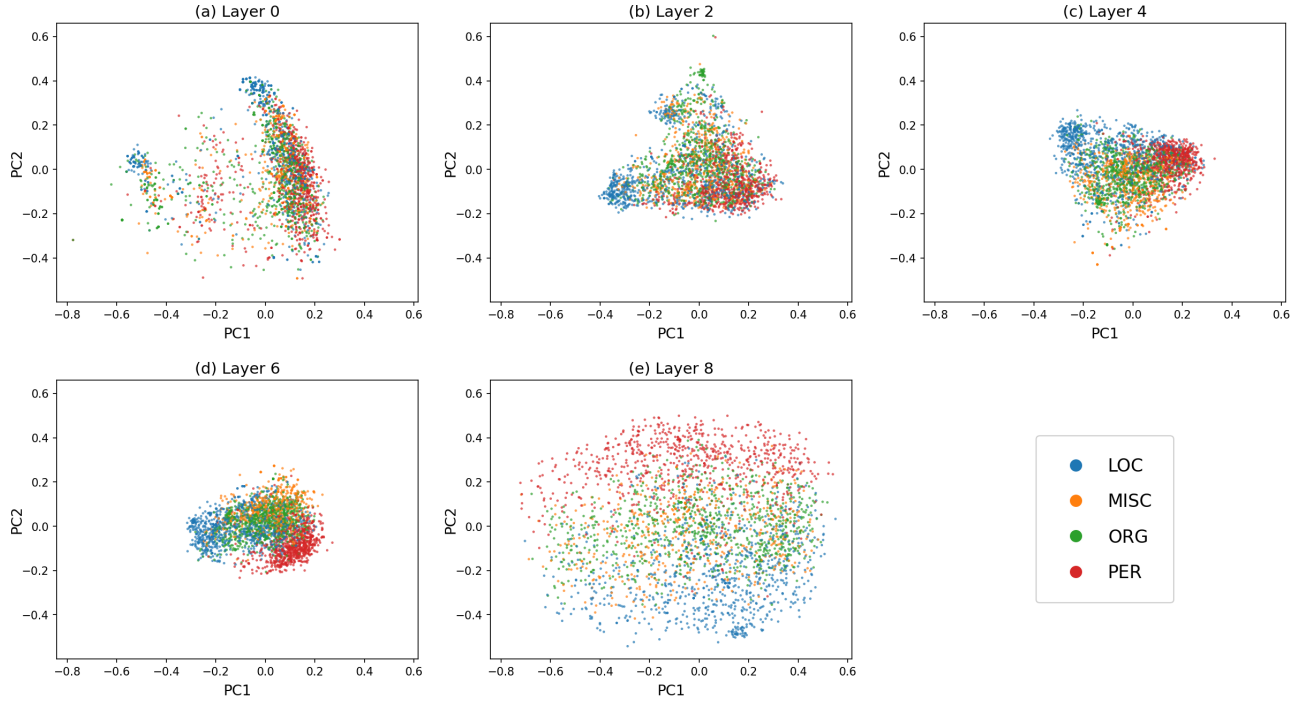
Figure 3. Layer-wise PCA projections of XLM-R representations coloured by entity type.



Note. Panels (a)–(e) display two-dimensional PCA projections of L2-normalised entity representations from layers 0, 3, 6, 9, and 12 of the XLM-R tagging model. Points are coloured by entity type (LOC, MISC, ORG, PER). L2 normalisation was applied before PCA to reduce the influence of differences in vector magnitude across layers. PCA was fitted separately for each layer; therefore, the axes should be interpreted as layer-specific visualisation axes rather than directly comparable dimensions.

Figure 3 shows a clear change in the distribution of XLM-R representations by entity type across layers. In the earlier layers, the four entity types overlap substantially, indicating that entity-type information is not yet strongly separated in the projected space. From layer 6 onward, the groups become more distinct, and by layers 9 and 12 the projections show much clearer separation between LOC, MISC, ORG, and PER. This visual pattern is consistent with the quantitative layer-wise results, which showed a strong increase in same-type kNN and entity-type silhouette in the deeper XLM-R layers. The figure therefore supports the interpretation that XLM-R develops increasingly type-oriented entity representations across layers.

Figure 4. Layer-wise PCA projections of mT5 entity representations coloured by entity type.



Note. Panels (a)–(e) show two-dimensional PCA projections of L2-normalised entity representations from layers 0, 2, 4, 6, and 8 of the mT5 generative model, coloured by entity type (LOC, MISC, ORG, PER). L2 normalisation was applied before PCA to reduce the influence of layer-specific differences in embedding magnitude. PCA was fitted separately for each layer; therefore, the axes should be interpreted as layer-specific visualisation axes rather than directly comparable dimensions across layers. The figure provides a qualitative view of how entity-type-related structure changes across layers.

Figure 4 shows a weaker layer-wise change in mT5 than was observed for XLM-R. The entity types are somewhat unevenly distributed in the early layers, but overlap remains substantial across all analysed layers. Later layers show some visible differentiation, especially between PER and the other entity types, but the separation between LOC, MISC, ORG, and PER remains less pronounced than in XLM-R. This visual pattern is consistent with the quantitative results in Table 5, where same-type kNN increases moderately but entity-type silhouette remains low across all analysed mT5 layers. Supplementary PCA and UMAP projections by language and entity type are provided in Appendix Figures 8–13. These figures support the qualitative layer-wise inspection but are not treated as primary evidence for the main claims.

In summary, the layer-wise results show that entity-type structure becomes much more pronounced across XLM-R layers, especially in the later layers. mT5 shows a weaker increase in same-type neighbourhood structure, but its global group separation remains limited.

Entity-type differences in clustering

This section examines whether the clustering patterns reported so far differ across entity types. To address SRQ3, it compares language-related clustering separately for LOC, MISC, ORG, and PER in the XLM-R tagging model and the mT5 generative model.

Table 6. Final-layer entity-conditioned language clustering metrics for XLM-R and mT5.

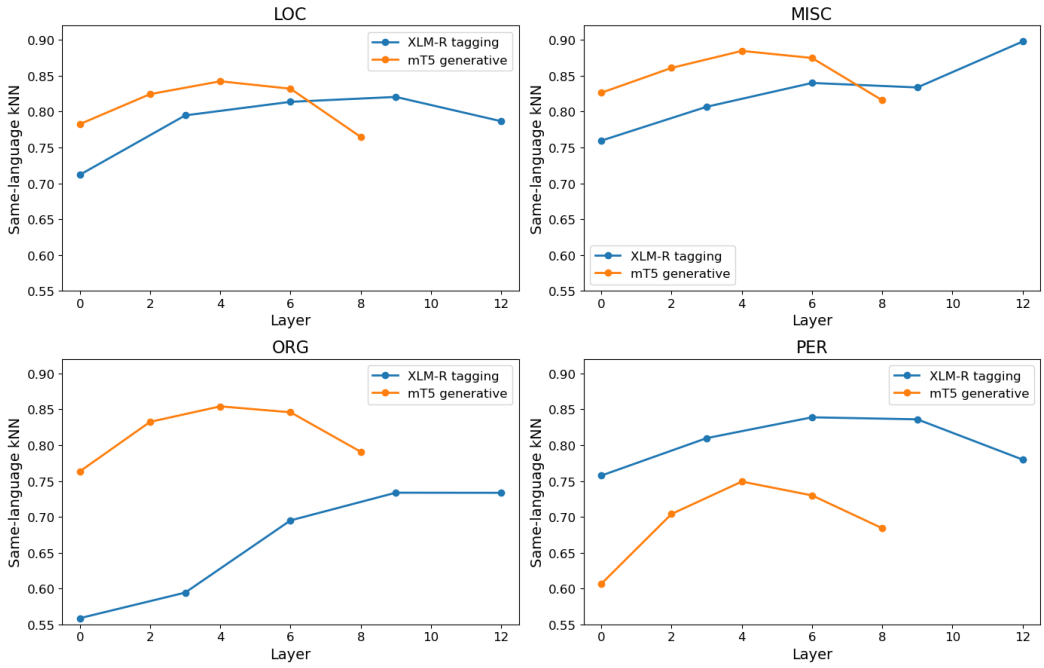
Model	Entity type	Same-language kNN	Language silhouette	N
XLM-R tagging	LOC	0.789	0.021	1183
	MISC	0.896	0.025	2532
	ORG	0.733	0.010	3391
	PER	0.781	0.007	2228
mT5 generative	LOC	0.766	0.012	3460
	MISC	0.803	0.018	2229
	ORG	0.785	0.017	3934
	PER	0.669	0.006	3380

Note. Same-language kNN measures the proportion of nearest neighbours within each entity type that share the same language. Language silhouette measures the degree of language-based separation within each entity type. *N* indicates the number of entity representations included for each entity type and model.

Table 6 reports language-conditioned clustering metrics separately for each entity type. In XLM-R, same-language kNN is highest for MISC (0.896), followed by LOC (0.789), PER (0.781), and ORG (0.733). In mT5, same-language kNN is also highest for MISC (0.803), followed by ORG (0.785), LOC (0.766), and PER (0.669). This indicates that residual language-based neighbourhood structure is not evenly distributed across entity categories. In both models, MISC shows the strongest tendency for entities to be surrounded by same-language neighbours, suggesting that this category is particularly language- or dataset-sensitive. By contrast, PER shows comparatively lower language-conditioned clustering in mT5, which may reflect the more cross-linguistically shared surface forms of many person names.

Language silhouette values remain low for all entity types, ranging from 0.007 to 0.025 in XLM-R and from 0.006 to 0.018 in mT5. This means that although same-language neighbours occur locally within entity types, the languages do not form clearly separated global clusters inside those entity categories. Overall, Table 6 suggests that cross-lingual alignment varies by entity type: language effects persist most strongly in local neighbourhoods, especially for MISC, but remain weak at the level of broader group separation.

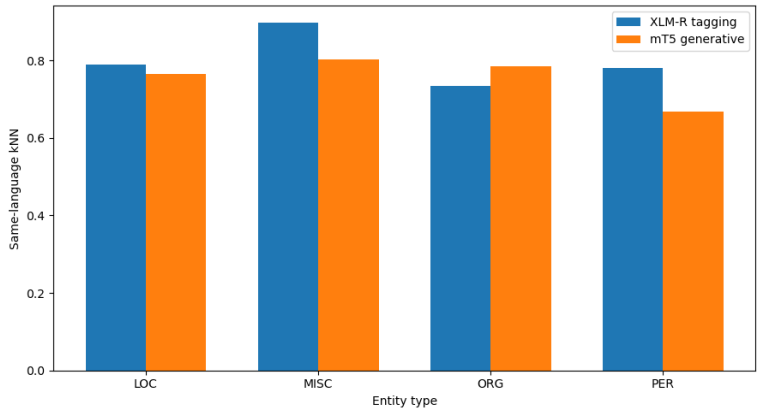
Figure 5. Layer-wise same-language kNN within entity types for XLM-R and mT5.



Note. Each panel shows same-language kNN across analysed layers for one entity type. Higher values indicate a larger proportion of nearest neighbours from the same language within that entity type. XLM-R layers are shown for layers 0, 3, 6, 9, and 12; mT5 layers are shown for layers 0, 2, 4, 6, and 8.

Figure 5 shows that same-language kNN varies across entity types and layers in both models. In XLM-R, values generally increase with depth for all four entity types, although the size of the increase differs across categories. In the final analysed XLM-R layer, MISC has the highest same-language kNN, while ORG has the lowest. In mT5, same-language kNN rises from the early to the middle layers for all four entity types and then declines in the final analysed layer. Across most analysed layers, mT5 shows higher same-language kNN than XLM-R for LOC, MISC, and ORG, whereas XLM-R shows higher values for PER.

Figure 6. Final-layer same-language kNN by entity type for XLM-R and mT5.



Note. Bars show final-layer same-language kNN for each entity type in the XLM-R tagging model and the mT5 generative model. Higher values indicate a larger proportion of nearest neighbours from the same language within the corresponding entity type.

Figure 6 summarises the final-layer same-language kNN values by entity type. In both models, MISC has the highest same-language kNN. The lowest value is observed for ORG in XLM-R and for PER in mT5. XLM-R has higher values than mT5 for LOC, MISC, and PER, whereas mT5 has a higher value for ORG.

Overall, the entity-type-conditioned results show that residual language structure is not uniform across entity categories. MISC shows the strongest same-language neighbourhood concentration in both models, while the lowest values occur for ORG in XLM-R and PER in mT5.

Model-paradigm differences

This section compares the representation geometry of the XLM-R tagging model and the mT5 generative model under the present experimental setups. The comparison focuses on differences in clustering metrics, layer-wise trends, projection-based visualisations, and entity-type-specific patterns.

Table 3 shows higher final-layer entity-type metrics for XLM-R than for mT5. In XLM-R, same-type kNN is 0.972 and entity-type silhouette is 0.703, whereas in mT5 the corresponding values are 0.829 and 0.067. Language silhouette remains low in both models. Table 4 shows a similar difference for cluster-recovery and centroid-distance metrics. In XLM-R, entity-type NMI and ARI are high, while language NMI and ARI are close to zero. The mean centroid distance is also larger for entity type than for language. In mT5, the entity-type values are lower, although they remain higher than the corresponding language-based values.

Figure 2 shows that the two models also differ in their layer-wise profiles. In XLM-R, same-type kNN and entity-type silhouette increase strongly across layers, especially in the later layers. In mT5, same-type kNN also increases, but more moderately, and entity-type silhouette remains low across all analysed layers. Same-language kNN increases in both models, but the difference between same-language and same-type metrics is larger in XLM-R.

The qualitative projections show a similar contrast. In Figure 1, the final analysed XLM-R layer shows clearer separation by entity type than by language. The same contrast is visible in mT5, but entity-type separation is less pronounced. Figures 3 and 4 further show that entity-type separation becomes clearer across layers in XLM-R, whereas the mT5 projections remain more overlapping.

Table 6 and Figures 5 and 6 show that the two models also differ across entity types. In the final analysed layer, XLM-R has higher same-language kNN than mT5 for LOC, MISC, and PER, whereas mT5 has a higher value for ORG. Taken together, the model comparison shows that the XLM-R tagging setup produces a sharper and more globally separated entity-type geometry than

the mT5 generative setup, while both models retain some degree of language-conditioned neighbourhood structure.

Exploratory attribution analysis

To complement the representation-geometric analysis, an exploratory attribution analysis was conducted using Integrated Gradients. The aim of this analysis was not to provide a full explainability benchmark, but to examine which parts of the input contributed most strongly to model predictions and whether the resulting attribution patterns were broadly consistent across the tagging and generative setups. For both models, attribution scores were aggregated over four predefined input regions: the target token, other tokens within the same entity span, the left context, and the right context.

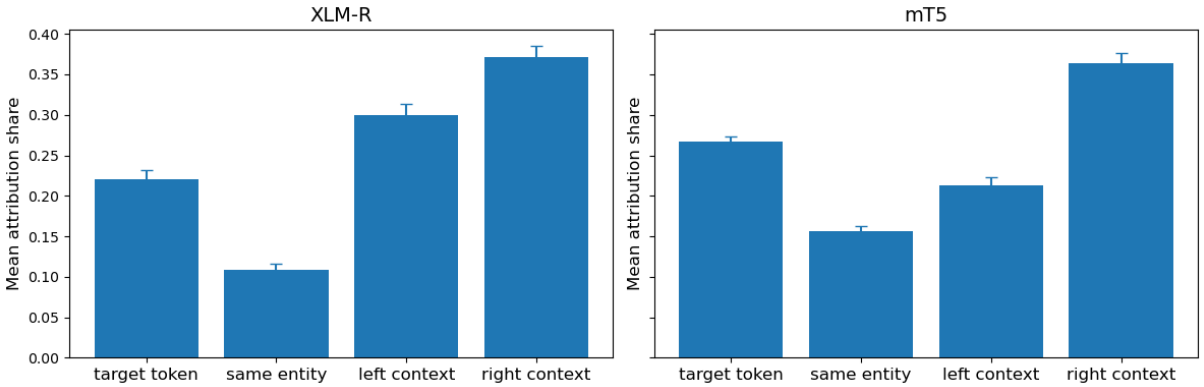
Table 7. Mean attribution share by model and input region with bootstrap confidence intervals.

Model	Input region	Mean attribution share	95% CI lower	95% CI upper
XLM-R tagging	Right context	0.372	0.358	0.386
	Left context	0.300	0.286	0.313
	Target token	0.220	0.209	0.232
	Same entity span	0.108	0.100	0.116
mT5 generative	Right context	0.364	0.351	0.377
	Left context	0.213	0.203	0.223
	Target token	0.267	0.260	0.273
	Same entity span	0.156	0.149	0.163

Note. Values report the mean share of total Integrated Gradients attribution assigned to each predefined input region. Confidence intervals are 95% bootstrap confidence intervals. “Same entity span” refers to other tokens belonging to the same entity mention as the target token.

For the XLM-R tagging model, attribution was concentrated more strongly in contextual regions than in the target token itself. As shown in Table 7 and Figure 7, right-context tokens received the largest average attribution share (M = 0.372, 95% CI [0.358, 0.386]), followed by left-context tokens (M = 0.300, 95% CI [0.286, 0.313]). The target token received a smaller share overall (M = 0.220, 95% CI [0.209, 0.232]), while tokens within the same entity span received the smallest share (M = 0.108, 95% CI [0.100, 0.116]).

Figure 7. Mean Integrated Gradients attribution share across input regions for XLM-R and mT5.



Note. Bars show the mean share of total Integrated Gradients attribution assigned to each input region. Error bars indicate bootstrap 95% confidence intervals. "Same entity" refers to other tokens within the same entity span as the target token.

A similar overall tendency was observed for the mT5 generative model, although the attribution setup is less direct in this case because the model produces decoder-side outputs rather than token-aligned source labels. Here too, right context received the largest average attribution share (M = 0.364, 95% CI [0.351, 0.377]). However, the remaining regional distribution differed somewhat from the tagging model. In mT5, the target token received the second-largest share (M = 0.267, 95% CI [0.260, 0.273]), followed by left context (M = 0.213, 95% CI [0.203, 0.223]), while the same-entity region again received the smallest share (M = 0.156, 95% CI [0.149, 0.163]).

Table 8. Paired Wilcoxon comparisons of attribution shares across input regions by model.

Model	Comparison	Mean difference	Wilcoxon statistic	Raw p	Corrected p	Significant
XLM-R tagging	Target token vs left context	-0.079	236567	7.76e-17	1.55e-16	True
	Target token vs right context	-0.151	166491	4.80e-48	1.92e-47	True
	Target token vs same entity span	0.112	173579	3.97e-44	1.19e-43	True
	Left context vs right context	-0.072	248021	2.16e-08	2.16e-08	True
mT5 generative	Target token vs left context	0.053	225275	4.22e-20	4.22e-20	True
	Target token vs right context	-0.097	213156	1.01e-24	2.02e-24	True
	Target token vs same entity span	0.110	51363	6.28e-135	2.51e-134	True
	Left context vs right context	-0.151	177347	2.52e-41	7.56e-41	True

Note. The table reports paired Wilcoxon signed-rank tests comparing attribution shares between predefined input regions within each model. Mean difference is computed as the first region minus the second region in each comparison. P-values are adjusted within each model using the Holm correction. "Same entity span" refers to other tokens within the same entity mention as the target token.

The paired Wilcoxon tests reported in Table 8 provide statistical comparisons between attribution regions within each model. For XLM-R, all four tested contrasts are statistically

significant after Holm correction. The target token receives less attribution than both left context and right context, but more than the same-entity region. Left and right context also differ significantly, with right context receiving the larger share.

The same applies to mT5: all reported region contrasts are statistically significant after Holm correction. In mT5, the target token receives more attribution than left context and the same-entity region, but less attribution than right context. Left and right context also differ significantly, with right context again receiving the larger share.

Overall, Tables 7 and 8 show that right context has the highest mean attribution share in both models, while the same-entity region has the lowest. The target-token share differs between the two models: it is lower than both context regions in XLM-R, but higher than left context in mT5.

Discussion

Summary of Key Findings

This thesis examined how named entity representations are organised in multilingual Transformer models across languages, layers, entity types, and modelling setups. The central finding is that the analysed representations are structured more strongly by entity type than by language. In both XLM-R and mT5, final-layer entity representations showed higher type-based than language-based clustering. However, this pattern was substantially clearer in XLM-R than in mT5.

A second key finding is that representation geometry changes across layers. In XLM-R, entity-type structure became increasingly pronounced in later layers, while mT5 showed a weaker layer-wise development. This indicates that cross-lingual semantic organisation is not static across representational depth and differs between the two implemented model setups.

The results also show that multilingual entity representations are not fully language-neutral. By residual language structure, I refer to local language-conditioned patterns that remain even when entity type is the stronger global organising principle. In other words, languages do not form clearly separated clusters overall, but nearby neighbours of an entity are still often from the same language. This is visible in the combination of low language silhouette scores and relatively high same-language kNN values. MISC showed the strongest residual language-conditioned structure in both models, indicating that cross-lingual alignment varies across entity types.

Finally, the exploratory attribution analysis showed that both models assigned substantial attribution to contextual regions, especially right context. This suggests that entity predictions were associated not only with the target token or entity span itself, but also with surrounding contextual

cues. Because the attribution setup is more direct for XLM-R than for mT5, these findings should be interpreted as a behavioural complement to the geometric analysis rather than as a fully parallel explainability comparison.

Overall, the findings support the main claim of the thesis in a qualified form: in the analysed English, Spanish, and Dutch CoNLL setting, multilingual NER models form partially language-invariant semantic entity spaces. At the same time, this organisation is incomplete, layered, entity-type-dependent, and shaped by the modelling setup.

Why does semantic alignment increase across layers?

One possible explanation for the layer-wise increase in entity-type structure is that different layers encode different levels of abstraction. Earlier Transformer layers are often more closely tied to lexical, subword, and local contextual information, whereas later layers are more strongly shaped by task-relevant distinctions. In the present thesis, this means that early representations may still preserve more surface-level and language-conditioned information, while later layers increasingly reflect the semantic categories required for NER.

This interpretation is especially plausible for XLM-R, where fine-tuning directly optimises token-level BIO predictions. The model is trained to assign entity labels to contextual token representations, so later layers are likely to become increasingly organised around distinctions such as PER, LOC, ORG, and MISC. The observed increase in entity-type clustering across XLM-R layers may therefore reflect the combined effect of multilingual pretraining and supervised NER fine-tuning: multilingual pretraining provides partially shared cross-lingual representations, while task-specific fine-tuning sharpens the entity-type structure in the upper layers.

At the same time, the persistence of same-language neighbourhood structure shows that this process does not remove language-specific information completely. Instead, the layer-wise results suggest a gradual reweighting of representational structure: semantic entity-type information becomes more dominant in later layers, but residual language-conditioned structure remains locally visible. This supports a graded view of cross-lingual alignment rather than a view in which deeper layers become fully language-neutral.

Why is XLM-R more entity-oriented than mT5?

The stronger entity-type organisation in XLM-R compared with mT5 may be related to the closer match between sequence tagging and the structure of the NER task. In the XLM-R setup, each supervised token position is directly trained to predict a BIO label. This creates an explicit connection between contextual token representations and entity categories. As a result, the final

hidden states are likely to become more separable according to the labels required by the classification head.

In the mT5 setup, the learning signal is less directly attached to individual source tokens. The model receives an input sentence and generates a linearised target sequence containing entity spans and labels. The entity-type decision is therefore mediated through the encoder–decoder architecture and the generation process rather than being represented as a direct token-level classification problem. This may make entity-category information less sharply localised in the encoder-side span embeddings extracted for the geometric analysis. In other words, mT5 may still learn entity information, but that information may be distributed across encoder representations, decoder states, and output-generation dynamics rather than concentrated in the same way as in a token-classification model.

A second possible factor is predictive performance. XLM-R also performs better on the NER task than mT5 in the present experiments, and stronger task performance may be associated with cleaner task-relevant representation geometry. However, this explanation should be treated cautiously, because the present study does not include controlled experiments that separate performance, architecture, task formulation, and embedding-extraction method. The difference between XLM-R and mT5 should therefore be interpreted as a difference between two implemented modelling setups rather than as evidence that encoder-only models are generally superior to encoder–decoder models for multilingual entity representation.

Theoretical Implications

The findings of this thesis support a view of the analysed multilingual Transformer representations as partially language-invariant rather than fully language-neutral. Across both models, entity representations were organised more strongly by semantic category than by language, yet the same-language kNN results show that language-related structure remained locally visible. In other words, entities did not form clearly separated global language clusters, as reflected in the low language silhouette scores in Table 3, but their nearest neighbours still often came from the same language. This pattern was especially visible in the entity-type-conditioned analysis in Table 6, where same-language neighbourhood concentration varied across PER, ORG, LOC, and MISC.

This indicates that multilingual NER does not require a perfectly shared interlingual space under these conditions. Instead, the results are more consistent with representational spaces in which shared semantic organisation coexists with residual language-conditioned structure. In theoretical terms, this supports a graded view of cross-lingual alignment: the analysed models

develop common semantic regions that are compatible with cross-lingual transfer while still preserving traces of language-specific encoding.

The additional KMeans and centroid-distance results sharpen this interpretation by showing that stronger entity-type structure is visible not only in neighbourhood purity and silhouette-based separation, but also in unsupervised cluster recovery and group-level geometry. In other words, entity-type organisation is not limited to local neighbourhood relations or projection-based visualisation but appears across multiple levels of geometric analysis. This strengthens the claim that, for the analysed models, multilingual named entity representations are not merely mixed across languages in a loose way, but can show semantically meaningful global structure, particularly in the XLM-R tagging setup.

A second implication concerns the idea of a shared multilingual embedding space. The results do not suggest that named entity representations collapse into a single undifferentiated semantic space across languages. Rather, they point to a more structured configuration in which entity type is the stronger organising principle, while language remains visible in local neighbourhood patterns. This means that entities do not form clearly separated global language clusters, as indicated by the low language silhouette scores in Table 3, but their nearest neighbours still often come from the same language, as shown by the same-language kNN values. This language-related neighbourhood structure also differs by entity category: in the entity-type-conditioned analysis in Table 6, MISC shows the strongest same-language concentration in both models. These findings therefore support theoretical accounts of multilingual representation learning that treat shared space as a matter of degree and internal structure rather than as a binary presence or absence of alignment.

A third implication concerns layer-wise organisation. The strong increase in entity-type clustering across XLM-R layers suggests that semantic abstraction is not uniformly present across representational depth but becomes more visible in later layers of the fine-tuned model. This is consistent with accounts of Transformer representations as hierarchically organised, with later layers often reflecting more task-relevant distinctions. At the same time, the weaker layer-wise consolidation observed in mT5 suggests that cross-lingual semantic structure does not emerge equally clearly across all modelling setups. Without a non-fine-tuned baseline, the analysis cannot fully separate the effects of multilingual pretraining from those of NER fine-tuning, but it does show that the analysed representation structure changes substantially across layers.

This leads to a broader implication for multilingual NER itself. The contrast between the XLM-R tagging model and the mT5 generative model suggests that the implemented modelling setup matters for representation geometry. Under the present experimental conditions, the tagging setup produced a much more sharply structured entity-type space than the generative setup.

However, this contrast should be interpreted cautiously. The two models differ not only in architecture, but also in task formulation, output structure, optimisation dynamics, and the degree of methodological access available for analysis. The present comparison therefore does not isolate architecture alone as the source of the observed differences. Even so, the results indicate that different multilingual NER setups may lead to substantially different degrees of semantic organisation in the analysed representation space.

The comparison between XLM-R and mT5 also suggests that multilingual entity spaces are shaped by task formulation. In the present experiments, the sequence-tagging setup produced a more sharply separated entity-type space than the generative setup. This may be because BIO-based token classification provides a more direct learning signal between input-token representations and entity categories, whereas the generative formulation mediates entity prediction through an encoder–decoder output sequence. The implication is that shared semantic structure in multilingual NER is not only a consequence of multilingual pretraining but also depends on how the NER task is operationalised during fine-tuning. However, because the two setups differ in architecture, output format, and embedding-extraction procedure, this contrast should be interpreted as a property of the implemented models rather than as a general ranking of tagging and generative approaches.

The exploratory attribution results add a behavioural dimension to these implications. As shown in Table 7, right context received the highest mean attribution share in both models, while the same-entity region received the lowest share. In XLM-R, right context accounted for 0.372 of the attribution share, compared with 0.108 for the same-entity region. In mT5, right context likewise received the highest share at 0.364, while the same-entity region received 0.156. The paired Wilcoxon comparisons in Table 8 further show that these regional differences were statistically significant after Holm correction. These results indicate that attribution was not concentrated only on the target token or entity span itself, but also on surrounding contextual cues. At the same time, the two models differed in the relative contribution of the target token and left context, suggesting that they rely on contextual information in partly different ways. More cautiously, these findings suggest that the analysed predictions are not explained purely by entity-local evidence but also involve contextual cues around the target.

Taken together, the results suggest that multilingual NER representation geometry cannot be explained by semantic alignment alone. Entity type is the strongest organising principle, but language, task formulation, and contextual cue use continue to shape how this structure appears in the analysed models. The findings therefore support a graded account of multilingual representation learning: shared semantic structure emerges, but it remains partial, layered, and dependent on the modelling setup.

Comparison to Prior Work

The findings of this thesis are broadly consistent with prior work showing that multilingual representations are often partially shared rather than fully language-neutral. Earlier studies have argued that multilingual models do not form a perfectly unified semantic space but instead combine shared semantic structure with residual language-specific organisation. This view is reflected in work on mBERT and related models, which finds evidence for cross-lingual transfer and shared structure, but also clear limitations and language-pair-specific variation (Libovický et al., 2020; Pires et al., 2019). The present results align closely with that position. In both XLM-R and mT5, entity-type clustering was stronger than language-based clustering, yet same-language neighbourhood concentration remained substantial and language-conditioned variation persisted across entity types.

The present findings are also consistent with more recent work questioning strong dominant-language accounts of multilingual processing. Basirat (2025) argues that multilingual large language models are better characterised by coordinate learning, with only limited evidence for stronger compound-style integration. Shani and Basirat (2025) likewise report only weak evidence for a dominant-language explanation of multilingual behaviour and instead argue for a combination of language-specific and shared representational spaces. This is compatible with the present thesis, which finds stronger entity-type clustering alongside residual language structure rather than complete language-neutrality or clear dominance by one language. Although the current study focuses on multilingual NER rather than general multilingual language modelling, the results are consistent with the same broader view: multilingual structure appears to be graded and internally differentiated rather than reducible to a single dominant-language space.

The results also fit well with prior work on representation geometry in multilingual models. Earlier geometric accounts suggest that hidden-state structure can reveal relationships that are not visible from downstream accuracy alone, including the coexistence of language-sensitive and language-neutral dimensions and the gradual emergence of task-relevant structure across layers (Libovický et al., 2020). The present findings are consistent with this perspective. In XLM-R, entity-type clustering increased strongly across layers, supporting the view that task-relevant semantic structure can become more visible at later representational depths rather than being equally apparent across all layers. mT5 showed a weaker version of the same trend, suggesting that cross-lingual semantic organisation may vary across modelling setups. In that respect, the thesis adds evidence from this controlled multilingual NER setting that named entity representations are not only partially aligned, but also differently structured across the two analysed model setups.

The results also argue against an overly strong interpretation of multilingual alignment as full convergence into a language-agnostic shared space. Strong multilingual transfer or good downstream performance does not necessarily mean that language-specific structure disappears. In the present analysis, semantic clustering was stronger than language clustering, but same-language kNN remained relatively high, especially for categories such as MISC. This indicates that local language-conditioned grouping persisted even when entity type was the dominant organising principle. The results are therefore compatible with a representational space in which semantic organisation is stronger overall, but language remains locally visible. In this respect, the findings support a graded account of multilingual organisation rather than a binary view of alignment as either present or absent (Libovický et al., 2020; Pires et al., 2019).

The comparison between XLM-R and mT5 also qualifies the assumption that multilingual models trained on the same task necessarily develop similar internal geometry. While both models showed stronger type-based than language-based structure, the difference was much sharper in XLM-R than in mT5. This suggests that the implemented model setup may affect how cross-lingual semantic structure is realised internally. That conclusion is compatible with broader multilingual representation work, but it also adds a more specific comparison between a tagging-based and a generative NER setup. In the present results, alignment is not simply present or absent; rather, its strength and form vary across the two analysed modelling setups. This is consistent with broader work showing that model design choices influence multilingual representation learning and transfer behaviour, including in large-scale multilingual encoders such as XLM-R (Conneau, Khandelwal, et al., 2020).

The exploratory attribution findings can also be related to prior work on interpretability. Integrated Gradients is an established attribution method, and gradient-based attribution has also been applied to encoder-decoder settings, including sequence-to-sequence models (He et al., 2019). In the present thesis, the attribution analysis complements the geometric findings by showing that high attribution mass is assigned to contextual regions rather than only to isolated entity tokens. At the same time, the attribution comparison is methodologically asymmetric: attribution is more direct in the XLM-R tagging setup than in the generative mT5 setup, where source-side attribution must be approximated heuristically. The attribution results therefore do not function as a strict parallel benchmark across paradigms, but they do suggest a pattern that is compatible with prior interpretability work: entity predictions in the analysed models are associated with contextual evidence around the target, not only with the target token itself.

Overall, the thesis is consistent with prior work in treating multilingual alignment as real but incomplete, and it reinforces the idea that semantic and language-specific structure coexist in

multilingual Transformer representations. At the same time, it adds a more fine-grained perspective from multilingual NER by showing that this balance varies across representational depth, entity category, model setup, and attribution patterns. In this way, the thesis contributes to the literature by moving from general claims about multilingual alignment toward a more specific account of how named entity representations are organised in the analysed multilingual NER setting.

Limitations

This thesis has several limitations that should be taken into account when interpreting the findings. First, the study is restricted to CoNLL-style NER datasets. English data are taken from CoNLL-2003 and Spanish and Dutch data from CoNLL-2002, all using a shared BIO label inventory with the four entity types PER, ORG, LOC, and MISC. This provides a relatively clean and controlled comparison setting, but it also narrows the empirical scope. The findings therefore apply most directly to standard newswire-style multilingual NER and may not generalise automatically to other domains, annotation schemes, or more diverse entity inventories.

Second, the analysis covers only three languages: English, Spanish, and Dutch. This allows for a controlled analysis of cross-lingual variation, but it remains a limited sample of the linguistic diversity relevant to multilingual NLP. All three languages are Indo-European, and two of them are relatively closely related. As a result, the thesis cannot determine whether the same geometric and attribution-related patterns would hold for typologically more distant languages, languages with different scripts, or lower-resource settings. The present results should therefore be understood as evidence from a specific multilingual configuration rather than as a universal account of multilingual NER representation geometry.

Third, the modelling comparison is limited to two specific setups: the XLM-R tagging model and the mT5 generative model. These were chosen because they instantiate two important paradigms in multilingual NER, namely sequence tagging and generative prediction. However, this also means that the comparison does not isolate architecture alone as the source of the observed differences. The two setups differ simultaneously in architecture, task formulation, output structure, optimisation dynamics, and the degree of methodological access available for analysis. The findings should therefore be interpreted as evidence about these particular implemented setups rather than as a general comparison of encoder-only and encoder-decoder models or of multilingual NER architectures more broadly.

A related limitation is that the comparison does not include a decoder-only multilingual language model, such as mGPT or BLOOM. Including such a model would make it possible to extend the architectural comparison beyond the encoder-only and encoder–decoder setups analysed

here, and to examine whether causal language modelling produces different cross-lingual entity representation geometry. However, this was outside the scope of the present thesis due to computational constraints. Future work could therefore incorporate a decoder-only multilingual model to test whether the patterns observed for XLM-R and mT5 generalise to a third modelling paradigm.

A further limitation concerns the use of low-dimensional visualisation methods, especially UMAP. In this thesis, PCA and UMAP are used as exploratory tools to inspect geometric structure, while the main quantitative claims rely on metrics such as kNN purity, silhouette scores, cluster-recovery measures, and centroid-based comparisons. Even so, nonlinear projections such as UMAP can distort global relationships and may exaggerate or compress apparent cluster separation. For this reason, the visualisations should not be interpreted as exact maps of the original embedding space, but only as heuristic aids to interpretation. The quantitative metrics are therefore given greater weight than the visual impressions alone.

Another limitation concerns the construction of entity representations and the absence of additional geometry-specific baselines. Entity embeddings are derived from token-level hidden states using span-level mean pooling. This is a reasonable and stable choice for entity-level analysis, but it remains a modelling decision that reflects a particular operational choice of what an entity representation is. Mean pooling may smooth over internal structure within multi-token entities, and different pooling strategies, such as first-token pooling or max pooling, could in principle produce somewhat different geometric patterns. More broadly, the thesis does not systematically compare the analysed models with non-fine-tuned pretrained representations, random-feature baselines, monolingual models, or alternative pooling strategies. As a result, the analysis cannot fully separate which parts of the observed geometry are inherited from multilingual pretraining, induced by NER fine-tuning, or shaped by the span-pooling method. The layer-wise analysis partially addresses this issue by showing how structure changes across representational depth, but it does not replace a true baseline comparison. The results are best understood as evidence about the geometry of the implemented fine-tuned models under the chosen extraction procedure, rather than as a complete causal account of where that geometry originates.

The exploratory attribution analysis also has important limitations. In the XLM-R tagging setup, attribution can be linked directly to token-level label predictions, which makes the attribution signal comparatively interpretable. In the mT5 generative setup, however, attribution is less direct because predictions are produced on the decoder side rather than as token-aligned source labels. The source-side attribution analysis for mT5 therefore relies on a simplified heuristic decomposition rather than on a fully token-aligned explanation procedure. As a result, the attribution findings are

not equally strong across the two model setups and should not be interpreted as a fully symmetric explainability comparison. More generally, the attribution analysis is exploratory and serves as a behavioural complement to the geometric results rather than as a complete explainability evaluation. The explainability component is also limited to Integrated Gradients. Other interpretability methods, such as attention visualisation or SHAP, were considered during the design stage but were not included in the final experimental pipeline. The attribution results should therefore be understood as one exploratory perspective on model behaviour rather than as a comprehensive explainability analysis.

A further limitation concerns the behavioural scope of the thesis more generally. Although an error taxonomy was defined during the project design, the final thesis does not include a full quantitative error-analysis component comparing the distribution of boundary errors, label confusions, missed entities, and spurious entities across languages and model setups. The behavioural extension of the study therefore remains focused primarily on attribution patterns rather than on a broader systematic comparison of prediction error types. Future work could extend the present analysis by integrating a fuller error-analysis pipeline alongside the geometric and attribution-based results, for example by comparing error-type distributions across languages, entity categories, and model setups.

Viewed together, these limitations do not invalidate the results, but they do define their scope. The thesis provides evidence about multilingual named entity representation geometry in a controlled three-language CoNLL setting, using two specific multilingual Transformer-based modelling setups, a particular span-level embedding design, and an exploratory attribution-based behavioural component. Future work could strengthen the generality of the findings by extending the analysis to more languages, additional domains and datasets, alternative model families, different span-representation methods, fuller quantitative error analysis, and more direct attribution procedures for generative multilingual NER.

Conclusion

This thesis examined how named entity representations are organised in multilingual Transformer models and to what extent they form shared cross-lingual semantic spaces rather than language-specific clusters.

The main finding is that, in the analysed English, Spanish, and Dutch CoNLL setting, named entity representations are structured more strongly by entity type than by language, although this semantic organisation remains partial rather than fully language-neutral. In both XLM-R and mT5,

type-based clustering was stronger than language-based clustering in the final analysed layers, with the contrast being substantially clearer in the XLM-R tagging model. These findings support the central claim of the thesis that multilingual NER models form partially language-invariant semantic clusters rather than purely language-specific representational spaces.

A second conclusion is that multilingual representation geometry develops across layers rather than remaining static throughout the network. In XLM-R, semantic organisation became progressively stronger in later layers, as reflected in increasing same-type neighbourhood concentration, higher entity-type silhouette scores, and clearer separation in the PCA projections. mT5 showed a weaker version of the same tendency. This suggests that cross-lingual semantic organisation is not simply present from the outset but is consolidated through representational depth and shaped by both architecture and task formulation.

The results also show that residual language structure persists even where semantic clustering dominates. Same-language neighbourhood concentration remained substantial in both models, and the degree of language-conditioned structure varied across entity types. Multilingual entity spaces are therefore best understood as semantically organised but still language-conditioned, rather than as fully merged interlingual spaces. The comparison between XLM-R and mT5 indicates that the two modelling setups differ substantially in how clearly semantic structure appears in the analysed representation spaces. Although both models showed stronger type-based than language-based structure, the tagging setup produced a much sharper and more stable entity-type geometry than the generative setup under the present experimental conditions.

Finally, the exploratory attribution analysis added a behavioural perspective by showing that prediction decisions in both models depended strongly on contextual information, especially right context, rather than on the focal token alone. This complements the geometric analysis by indicating that semantically organised representations are operationalised through strongly context-conditioned decisions. Taken together, the findings suggest that multilingual named entity recognition is best understood as the interaction of semantic abstraction, residual language structure, representational depth, modelling paradigm, and contextual cue use.

Overall, the thesis provides evidence that, in this controlled English, Spanish, and Dutch CoNLL setting, the analysed Transformer models form shared semantic structure for named entities, but that this structure is incomplete, layered, and dependent on the implemented modelling setup. The findings therefore support a qualified view of multilingual representation learning in this setting: not full language invariance, but partial cross-lingual semantic alignment with persistent language-specific traces.

Bibliography

Basirat, A. (2025). Multilingual Learning Strategies in Multilingual Large Language Models. *Proceedings of the 5th Workshop on Multilingual Representation Learning (MRL 2025)*, (pp. 507-518).

Chai, Y., Liang, Y., & Duan, N. (2022). Cross-Lingual Ability of Multilingual Masked Language Models: A Study of Language Structure. In S. Muresan, P. Nakov, & A. Villavicencio (Eds), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 4702–4712). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.322

Chang, T. A., Tu, Z., & Bergen, B. K. (2022). The Geometry of Multilingual Language Model Representations. In Y. Goldberg, Z. Kozareva, & Y. Zhang (Eds), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (pp. 119–136). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.emnlp-main.9

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised Cross-lingual Representation Learning at Scale. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 8440–8451). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.747

Conneau, A., Wu, S., Li, H., Zettlemoyer, L., & Stoyanov, V. (2020). Emerging Cross-lingual Structure in Pretrained Language Models. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 6022–6034). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.536

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In J. Burstein, C. Doran, & T.

Solorio (Eds), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 4171–4186). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Doğruöz, A. S., & Sitaram, S. (2022). Language Technologies for Low Resource Languages: Sociolinguistic and Multilingual Insights. In M. Melero, S. Sakti, & C. Soria (Eds), *Proceedings of the 1st Annual Meeting of the ELRA/ISCA Special Interest Group on Under-Resourced Languages* (pp. 92–97). European Language Resources Association. https://aclanthology.org/2022.sigul-1.12/

European Commission. (2024, October 28). *Language Technologies and Multilingualism*. Shaping Europe's Digital Future. https://digital-strategy.ec.europa.eu/en/policies/language-technologies-multilingualism

Hämmerl, K., Libovický, J., & Fraser, A. (2024). Understanding Cross-Lingual Alignment—A Survey. *Findings of the Association for Computational Linguistics ACL 2024*, 10922–10943. https://doi.org/10.18653/v1/2024.findings-acl.649

He, S., Tu, Z., Wang, X., Wang, L., Lyu, M., & Shi, S. (2019). Towards Understanding Neural Machine Translation with Word Importance. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 953–962). Association for Computational Linguistics. https://doi.org/10.18653/v1/D19-1088

Hedderich, M. A., Lange, L., Adel, H., Strötgen, J., & Klakow, D. (2021). A Survey on Recent Approaches for Natural Language Processing in Low-Resource Scenarios. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational*

Linguistics: Human Language Technologies, 2545–2568. https://doi.org/10.18653/v1/2021.naacl-main.201

Hewitt, J., & Manning, C. D. (2019). A Structural Probe for Finding Syntax in Word Representations. In J. Burstein, C. Doran, & T. Solorio (Eds), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 4129–4138). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1419

Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, *2*(1), 193–218. https://doi.org/10.1007/BF01908075

Jain, A., Paranjape, B., & Lipton, Z. C. (2019). Entity Projection via Machine Translation for Cross-Lingual NER. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)* (pp. 1083–1092). Association for Computational Linguistics. https://doi.org/10.18653/v1/D19-1100

Jawahar, G., Sagot, B., & Seddah, D. (2019). What Does BERT Learn about the Structure of Language? In A. Korhonen, D. Traum, & L. Màrquez (Eds), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 3651–3657). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1356

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *374*(2065), 20150202. https://doi.org/10.1098/rsta.2015.0202

Li, B., Zhou, H., He, J., Wang, M., Yang, Y., & Li, L. (2020). On the Sentence Embeddings from Pre-trained Language Models. In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*

(pp. 9119–9130). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.emnlp-main.733

Li, J., Sun, A., Han, J., & Li, C. (2022). A Survey on Deep Learning for Named Entity Recognition. *IEEE Transactions on Knowledge and Data Engineering*, *34*(1), 50–70. https://doi.org/10.1109/TKDE.2020.2981314

Libovický, J., Rosa, R., & Fraser, A. (2020). On the Language Neutrality of Pre-trained Multilingual Representations. In T. Cohn, Y. He, & Y. Liu (Eds), *Findings of the Association for Computational Linguistics: EMNLP 2020* (pp. 1663–1674). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.findings-emnlp.150

McInnes, L., Healy, J., & Melville, J. (2020). *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction* (arXiv:1802.03426). arXiv. https://doi.org/10.48550/arXiv.1802.03426

McInnes, L., Healy, J., Saul, N., & Großberger, L. (2018). UMAP: Uniform Manifold Approximation and Projection. *Journal of Open Source Software*, *3*(29), 861. https://doi.org/10.21105/joss.00861

Mikolov, T., Chen, K., Corrado, G. s, & Dean, J. (2013). Efficient Estimation of Word Representations in Vector Space. *Proceedings of Workshop at ICLR*, *2013*.

Mousi, B., Durrani, N., Dalvi, F., Hawasly, M., & Abdelali, A. (2024). Exploring Alignment in Shared Cross-lingual Spaces. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 6326–6348. https://doi.org/10.18653/v1/2024.acl-long.344

Mueller, D., Andrews, N., & Dredze, M. (2020). Sources of Transfer in Multilingual Named Entity Recognition. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 8093–8104). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.720

Muller, B., Elazar, Y., Sagot, B., & Seddah, D. (2021). First Align, then Predict: Understanding the Cross-Lingual Ability of Multilingual BERT. In P. Merlo, J. Tiedemann, & R. Tsarfaty (Eds), *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume* (pp. 2214–2231). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.eacl-main.189

Nadeau, D., & Sekine, S. (2007). A survey of named entity recognition and classification. *Lingvisticae Investigationes*, *30*(1), 3–26. https://doi.org/10.1075/li.30.1.03nad

Pai, L., Gao, W., Dong, W., Ai, L., Gong, Z., Huang, S., Zongsheng, L., Hoque, E., Hirschberg, J., & Zhang, Y. (2024). A Survey on Open Information Extraction from Rule-based Model to Large Language Model. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds), *Findings of the Association for Computational Linguistics: EMNLP 2024* (pp. 9586–9608). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-emnlp.560

Pires, T., Schlinger, E., & Garrette, D. (2019). How Multilingual is Multilingual BERT? In A. Korhonen, D. Traum, & L. Màrquez (Eds), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 4996–5001). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1493

Qin, L., Chen, Q., Zhou, Y., Chen, Z., Li, Y., Liao, L., Li, M., Che, W., & Yu, P. S. (2025). A survey of multilingual large language models. *Patterns*, *6*(1). https://doi.org/10.1016/j.patter.2024.101118

Rahimi, A., Li, Y., & Cohn, T. (2019). Massively Multilingual Transfer for NER. In A. Korhonen, D. Traum, & L. Màrquez (Eds), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 151–164). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1015

Ramshaw, L., & Marcus, M. (1995). Text Chunking using Transformation-Based Learning. *Third Workshop on Very Large Corpora*. https://aclanthology.org/W95-0107/

Rogers, A., Kovaleva, O., & Rumshisky, A. (2020). A Primer in BERTology: What We Know About How BERT Works. *Transactions of the Association for Computational Linguistics*, *8*, 842–866. https://doi.org/10.1162/tacl_a_00349

Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, *20*, 53–65. https://doi.org/10.1016/0377-0427(87)90125-7

Ruder, S., Vulić, I., & Søgaard, A. (2019). A Survey of Cross-lingual Word Embedding Models. *Journal of Artificial Intelligence Research*, *65*, 569–631. https://doi.org/10.1613/jair.1.11640

Sang, E. T. K. (2002). Introduction to the CoNLL-2002 Shared Task: Language-Independent Named Entity Recognition. *COLING-02: The 6th Conference on Natural Language Learning 2002 (CoNLL-2002)*. CoNLL 2002. https://aclanthology.org/W02-2024/

Sang, E. T. K., & De Meulder, F. (2003). Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition. *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*, 142–147. https://aclanthology.org/W03-0419/

Schuster, T., Ram, O., Barzilay, R., & Globerson, A. (2019). Cross-Lingual Alignment of Contextual Word Embeddings, with Applications to Zero-shot Dependency Parsing. In J. Burstein, C. Doran, & T. Solorio (Eds), *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)* (pp. 1599–1613). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1162

Shah, C., Chandak, Y., Mane, A. M., Bergen, B., & Chang, T. A. (2024). Correlations between Multilingual Language Model Geometry and Crosslingual Transfer Performance. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, & N. Xue (Eds), *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources*

and Evaluation (LREC-COLING 2024) (pp. 4059–4066). ELRA and ICCL. https://aclanthology.org/2024.lrec-main.361/

Shani, N., & Basirat, A. (2025). Language Dominance in Multilingual Large Language Models. *Proceedings of the 8th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, (pp. 137-148).

Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic Attribution for Deep Networks. *Proceedings of the 34th International Conference on Machine Learning*, 3319–3328. https://proceedings.mlr.press/v70/sundararajan17a.html

Tanti, M., van der Plas, L., Borg, C., & Gatt, A. (2021). On the Language-specificity of Multilingual BERT and the Impact of Fine-tuning. In J. Bastings, Y. Belinkov, E. Dupoux, M. Giulianelli, D. Hupkes, Y. Pinter, & H. Sajjad (Eds), *Proceedings of the Fourth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP* (pp. 214–227). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.blackboxnlp-1.15

Tenney, I., Das, D., & Pavlick, E. (2019). BERT Rediscovers the Classical NLP Pipeline. In A. Korhonen, D. Traum, & L. Màrquez (Eds), *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 4593–4601). Association for Computational Linguistics. https://doi.org/10.18653/v1/P19-1452

Upadhyay, S., Faruqui, M., Dyer, C., & Roth, D. (2016). Cross-lingual Models of Word Embeddings: An Empirical Comparison. In K. Erk & N. A. Smith (Eds), *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 1661–1670). Association for Computational Linguistics. https://doi.org/10.18653/v1/P16-1157

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*.

Xie, J., Yang, Z., Neubig, G., Smith, N. A., & Carbonell, J. (2018). Neural Cross-Lingual Named Entity Recognition with Minimal Resources. In E. Riloff, D. Chiang, J. Hockenmaier, & J. Tsujii (Eds), *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing* (pp. 369–379). Association for Computational Linguistics. https://doi.org/10.18653/v1/D18-1034

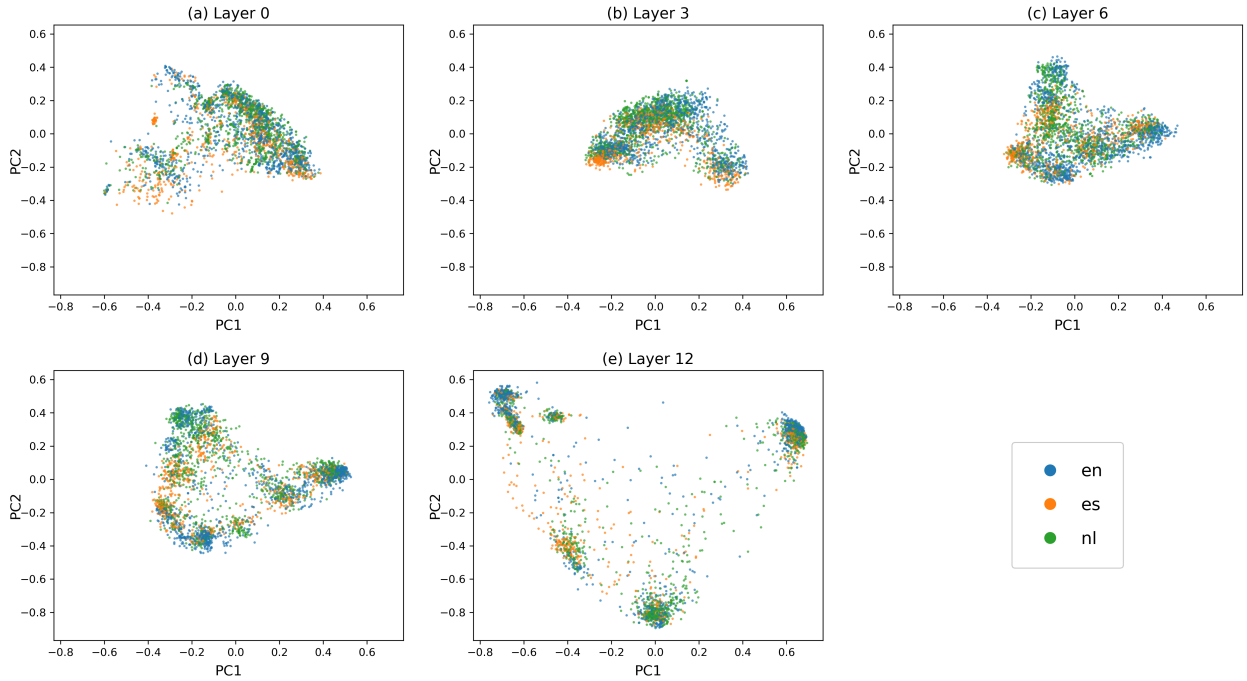
Yadav, D., Suravee, S., Strauß, T., & Yordanova, K. (2024). Cross-Lingual Named Entity Recognition for Low-Resource Languages: A Hindi-Nepali Case Study Using Multilingual BERT Models. In J. Sälevä & A. Owodunni (Eds), *Proceedings of the Fourth Workshop on Multilingual Representation Learning (MRL 2024)* (pp. 167–174). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.mrl-1.12

Yan, H., Gui, T., Dai, J., Guo, Q., Zhang, Z., & Qiu, X. (2021, June 2). *A Unified Generative Framework for Various NER Subtasks*. https://doi.org/10.48550/arXiv.2106.01223

Zhou, Y., & Srikumar, V. (2022). A Closer Look at How Fine-tuning Changes BERT. In S. Muresan, P. Nakov, & A. Villavicencio (Eds), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 1046–1061). Association for Computational Linguistics. https://doi.org/10.18653/v1/2022.acl-long.75

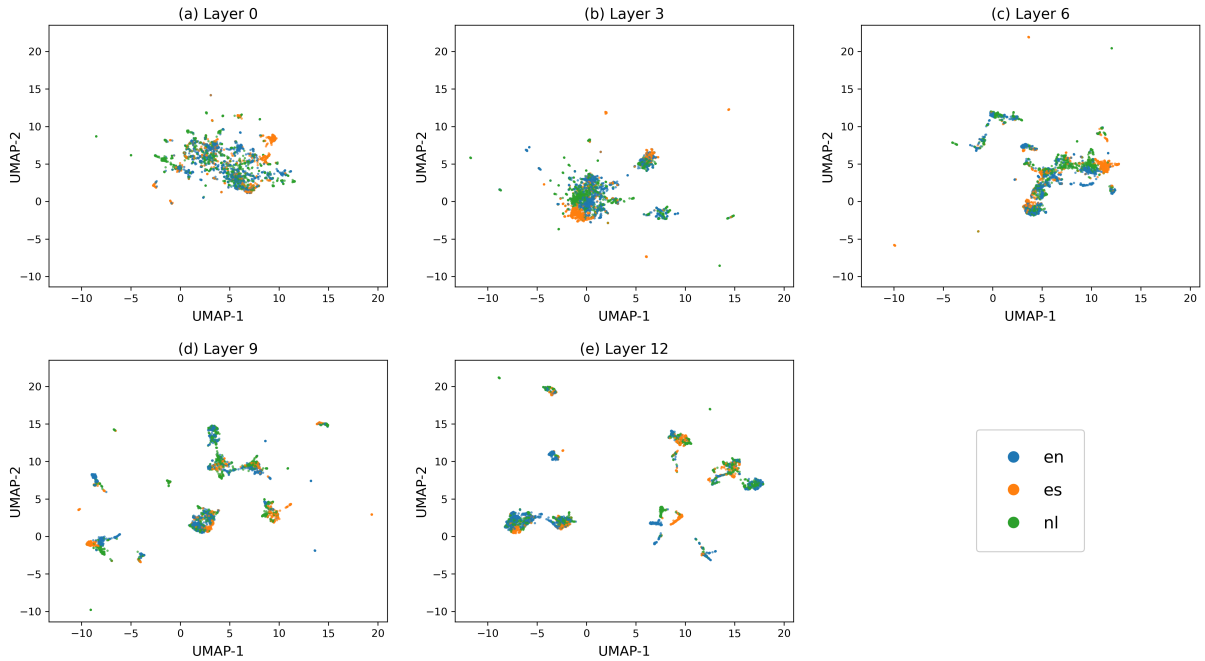
Appendix

Figure 8. Layer-wise PCA projections of XLM-R representations by language.



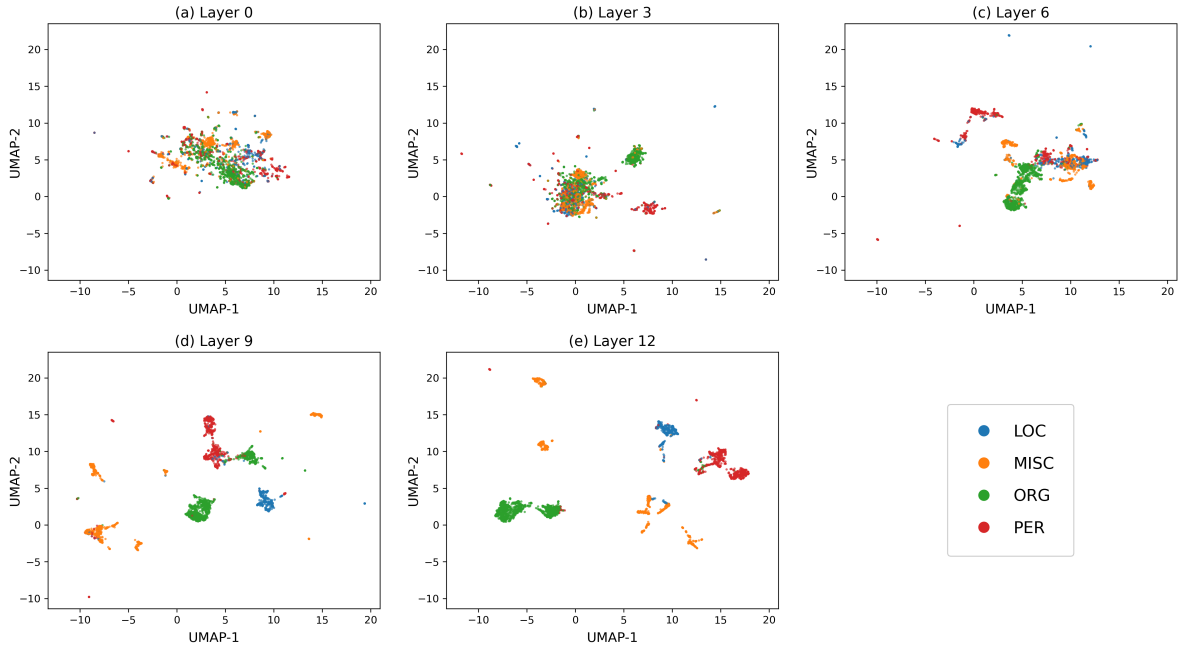
Note. Panels (a)–(e) show two-dimensional PCA projections of L2-normalised XLM-R entity representations from layers 0, 3, 6, 9, and 12, coloured by language: English (en), Spanish (es), and Dutch (nl). PCA was fitted separately for each layer, so the axes are layer-specific visualisation dimensions and should not be compared directly across panels.

Figure 9. Layer-wise UMAP projections of XLM-R representations by language.



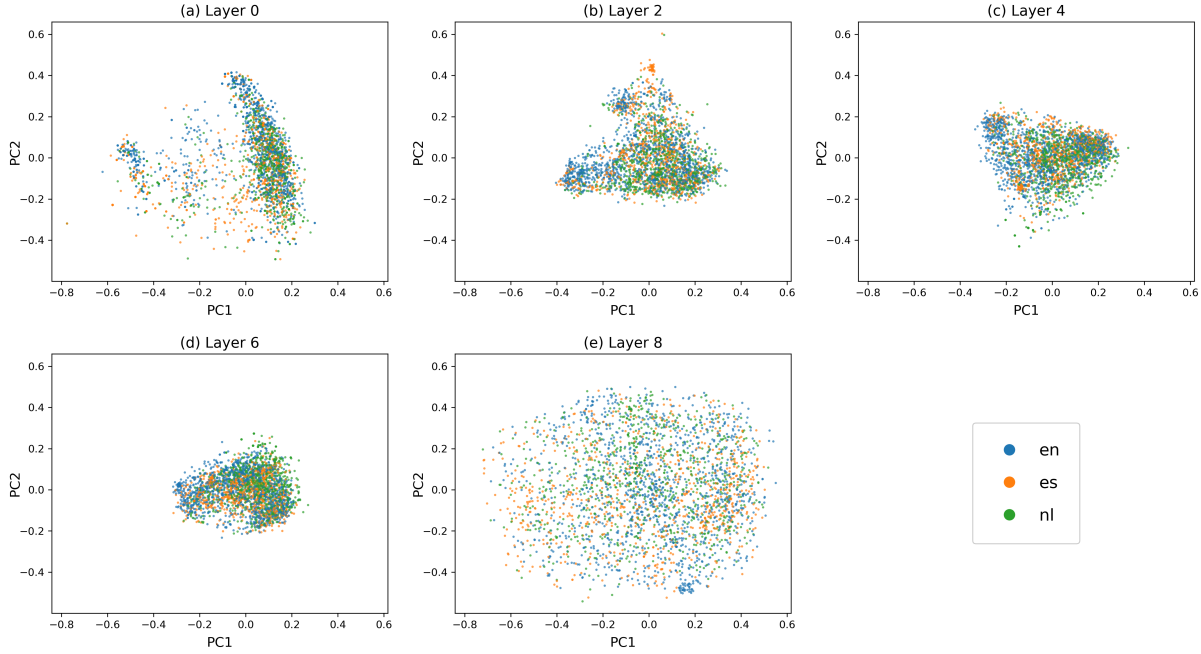
Note. Panels (a)–(e) show two-dimensional UMAP projections of XLM-R entity representations from layers 0, 3, 6, 9, and 12, coloured by language: English (en), Spanish (es), and Dutch (nl). UMAP was fitted separately for each layer, so the axes are layer-specific visualisation dimensions and should not be compared directly across panels.

Figure 10. Layer-wise UMAP projections of XLM-R representations by entity type.



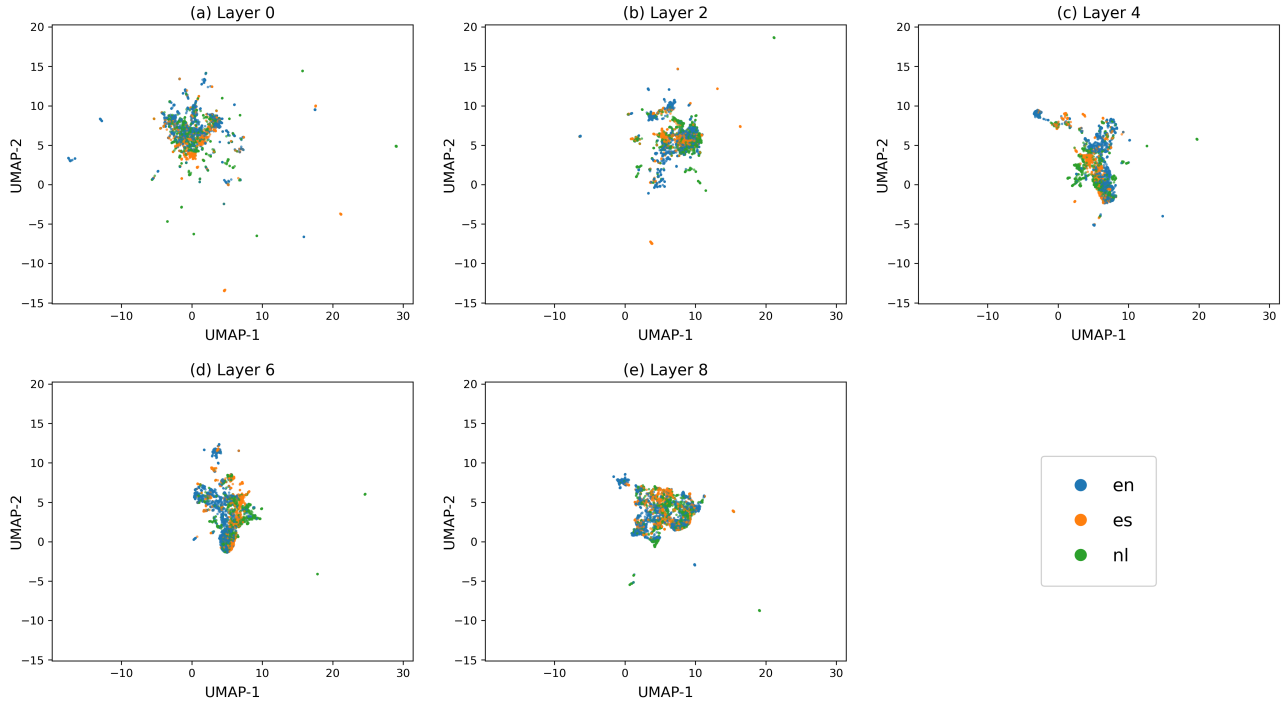
Note. Panels (a)–(e) show two-dimensional UMAP projections of XLM-R entity representations from layers 0, 3, 6, 9, and 12, coloured by entity type (LOC, MISC, ORG, PER). UMAP was fitted separately for each layer, so the axes are layer-specific and should not be compared directly across panels.

Figure 11. Layer-wise PCA projections of mT5 representations by language.



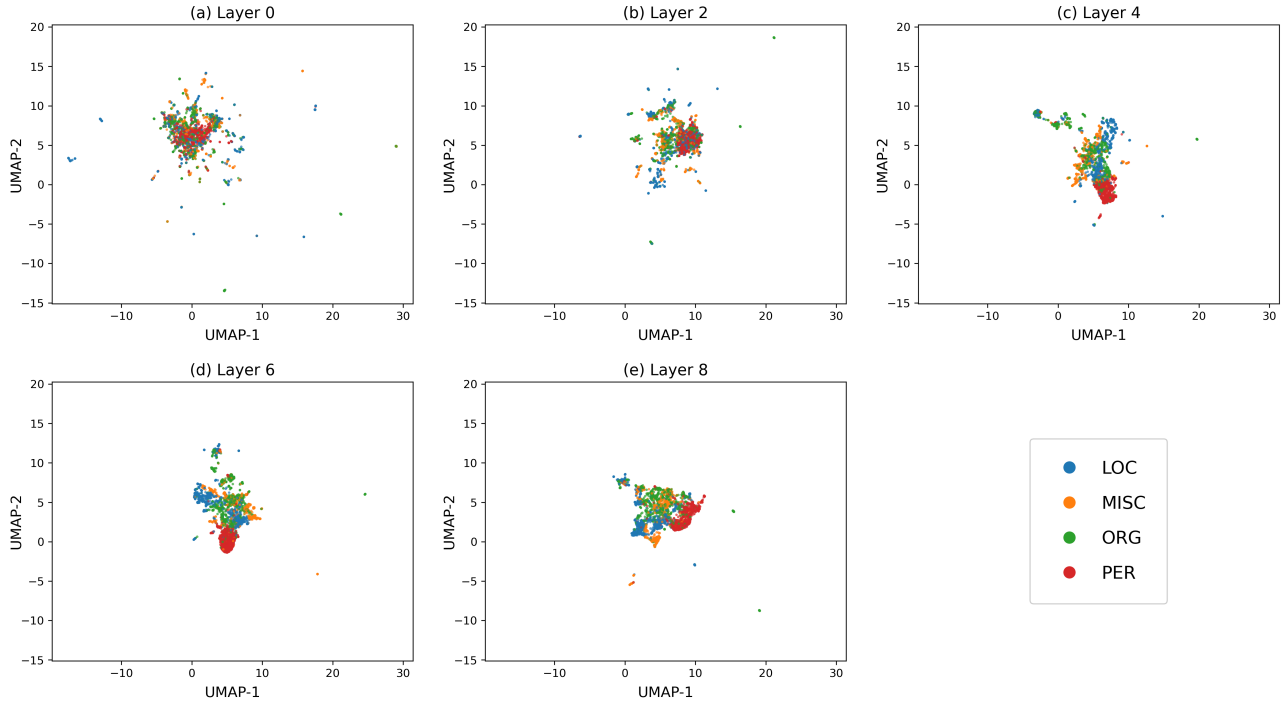
Note. Panels (a)–(e) show two-dimensional PCA projections of L2-normalised mT5 entity representations from layers 0, 2, 4, 6, and 8, coloured by language: English (en), Spanish (es), and Dutch (nl). PCA was fitted separately for each layer, so the axes are layer-specific and should not be compared directly across panels.

Figure 12. Layer-wise UMAP projections of mT5 representations by language.



Note. Panels (a)–(e) show two-dimensional UMAP projections of mT5 entity representations from layers 0, 2, 4, 6, and 8, coloured by language: English (en), Spanish (es), and Dutch (nl). UMAP was fitted separately for each layer, so the axes are layer-specific and should not be compared directly across panels.

Figure 13. Layer-wise UMAP projections of mT5 representations by entity type.



Note. Panels (a)–(e) show two-dimensional UMAP projections of mT5 entity representations from layers 0, 2, 4, 6, and 8, coloured by entity type (LOC, MISC, ORG, PER). UMAP was fitted separately for each layer, so the axes are layer-specific and should not be compared directly across panels.

Table 9. k-sensitivity analysis for final-layer kNN purity.

Model	Layer	k	Same-type kNN	Same-language kNN	Difference
XLM-R tagging	12	5	0.975	0.846	0.129
XLM-R tagging	12	10	0.972	0.798	0.174
XLM-R tagging	12	20	0.970	0.745	0.225
XLM-R tagging	12	50	0.967	0.674	0.293
mT5 generative	8	5	0.856	0.799	0.057
mT5 generative	8	10	0.829	0.760	0.069
mT5 generative	8	20	0.797	0.716	0.081
mT5 generative	8	50	0.749	0.659	0.090

Note. The table reports same-type and same-language kNN purity for k = 5, 10, 20, and 50. Across all neighbourhood sizes, same-type kNN remains higher than same-language kNN for both models.